

ENOUGH FOR EVERYONE

Limited by Biosphere, Institutions, Culture



Not Lacking Atoms

ENOUGH FOR EVERYONE

The Planet's Resources, the Limits of Abundance, and the Possible Road to a Post-Scarcity Economy

Humanity does not appear condemned by a shortage of atoms. It is constrained by the speed of transformation, by the biosphere, and by the institutions through which it decides who gains access to abundance.

Working edition based on public data available through August 28, 2026

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Author's Note

This work was created under the umbrella of Axiologic Research as part of two interconnected research programs, one dedicated to Artificial Intelligence and the other to Outfinitism, both led by Sînică Alboai. To explore the details of how these two programs intersect and build upon each other, we invite readers to visit the Outfinity Venture Studio page at www.outfinity.ch.

While Artificial Intelligence language models were used to assist with text generation, the foundational philosophy, thematic structure, and analytical approach derive exclusively from the author's decades of dedicated study of social phenomena, social technologies, and genuine hard technologies resulting from various European and self funded research projects. Recently, within the Achilles research project, we have been deeply studying AI generated literature and its automated verification using neuro symbolic approaches; all the free books made available to the public are part of the suite of works we use to test our latest technologies.

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Note on Method

This book begins with a simple question that is easy to formulate badly: does humanity have enough resources for material deprivation to stop organizing social life? The question is not whether the planet contains infinite quantities of energy, water, metals, or food. It does not. The useful question is whether there is enough physical, technological, and industrial potential to guarantee every person a safe, comfortable, dignified life, without obtaining essentials consuming most of their time and without destabilizing the natural systems on which we depend.

The analysis separates three levels that are often conflated. The first is physical potential: how much energy reaches Earth, how much material exists in the crust, how much water can be treated, and what biological yields are possible. The second is industrial capacity: mines, refineries, factories, grids, ports, storage, robots, skills, logistics chains, and the time required to build them. The third is social access: who owns the infrastructure, who can pay, which services are guaranteed, which risks are externalized, and who sets priorities. Abundance at the first level can coexist with bottlenecks at the second and poverty at the third.

The data used in this book were checked through August 28, 2026. The principal sources are reports and databases from the United Nations, the International Energy Agency, IRENA, FAO, UNEP, USGS, the International Labour Organization, OECD, the International Federation of Robotics, and peer-reviewed scientific literature. Estimates marked as “author’s estimate” are not institutional forecasts and should not be read as statistical confidence intervals. They are synthetic scenarios constructed from available data, technological trends, industrial lead times, and political constraints.

The book uses the English term post-scarcity because the concept has accumulated a distinct intellectual history. I use it, however, in a deliberately operational sense. A society reaches a basic post-scarcity threshold when nutritious food, water, household energy, decent housing, basic health care, education, connectivity, and reasonable mobility are universally accessible, resilient, and ecologically sustainable, while their combined cost no longer forces most people to sell most of their lives merely to survive. This is a far more modest target than infinite resources and, precisely for that reason, a far more relevant one.

The chapter on biology and culture applies an additional discipline. It separates what is supported by anthropological comparison, archaeology, experimental psychology, and behavioral genetics from the author’s inferences about a future society. Heritability is treated as population- and environment-dependent statistical variation, not as individual destiny. Observations about motivation, status, and domination are presented as distributions and possible mechanisms, not as a single moral essence of the species.

Throughout the book, calculations are retained only when they clarify an order of magnitude. Symbolic notation that imitates precision without adding explanatory power has been removed.

When progress depends on energy, materials, infrastructure, institutions, and distribution, the text simply says that the service appears only when all indispensable components function; it does not turn this engineering commonplace into a formula.

1. The Book in Ten Minutes: Conclusions Before the Demonstration

1.1 The Verdict and the Material Inventory

With reasonable probability, humanity has enough energy, matter, water, and biological capacity for no human being to remain constrained by lack of food, decent housing, basic medical treatment, education, connectivity, or a reasonable degree of mobility. This conclusion assumes neither infinite resources nor miraculous technology. It assumes the expansion of technologies already known, several plausible advances in automation, and an institutional reorganization that converts productive capacity into effective access. In the restricted sense used here, a basic post-scarcity era is physically possible within this century.

Absolute post-scarcity, however, is impossible. A person's time, other people's attention, highly desired places, intact ecosystems, political power, status, unique objects, irreversible experiences, and consent-based relationships cannot be multiplied on demand. Even if energy, intelligence, and labor become extremely cheap, not everyone can be first, most admired, the decision-maker, or the partner of the same person. Economics does not disappear; it shifts from calories, kilowatts, and labor hours toward time, authenticity, risk, social position, and institutional control.

The right question is therefore not whether we can eliminate every form of scarcity, but whether we can detach survival and dignity from competition for resources. My answer is yes. We can construct a floor below which no one falls without pretending that every desire will be satisfied or every outcome made equal. Realistic abundance means universal sufficiency, redundancy, and the freedom to refuse degrading work; it does not mean unlimited consumption or the homogenization of lives.

The book's most important result is that the near-term limit does not appear to be the planet's raw inventory. The limits are the speed at which we open mines and refineries, build grids and housing, train people, automate processes, protect ecosystems, and change rules of ownership. These processes are slow, conflict-ridden, and unevenly distributed. A society can have cheap energy and poor citizens; enough food and hunger; productive robots and people with little bargaining power. Post-scarcity is not an automatic outcome of technology but a political achievement enabled by technology.

Energy is the most favorable part of the balance sheet. The world energy system today operates at roughly twenty terawatts, while Earth intercepts approximately 174,000 terawatts of solar radiation. We cannot and should not capture more than a tiny fraction of that flow. Even a fifty- or one-hundred-terawatt civilization would use only a very small share of incident solar energy. The real problem is industrial: panels, turbines, power plants, grids, transformers, storage, materials, permitting, maintenance, and the cost of capital. The rapid growth of renewables shows that supply

can be multiplied within a generation, but availability at every hour and in every region requires a plural system, not a single technology.

Nor is matter, in general, close to geological exhaustion. Iron, aluminum, silicon, and limestone are abundant, although extracting and processing them consumes energy, degrades ecosystems, and produces emissions. For copper, lithium, graphite, nickel, and other critical materials, the near-term danger is a capacity shortage: mining and refining projects are not expanding fast enough, and production is geographically concentrated. Recycling, design for long life, substitution, and standardization can reduce pressure, but they cannot fully replace extraction while global infrastructure is still expanding. We are not “running out of atoms”; for years or decades, we may run short of enough mines, refineries, permits, and logistics chains.

Food is already produced at a scale that could feed everyone, but not necessarily in the form, place, and at the price each person needs. Contemporary hunger coexists with enormous losses and waste, the use of grain for animals and fuels, conflict, and populations unable to afford a healthy diet. Higher yields, lower losses, plant breeding, efficient irrigation, precision agriculture, and a larger share of lower-impact proteins create substantial room for improvement. Vertical farming is useful for certain perishable crops and in certain cities, but it is not an energetically rational solution for staple grains. Land, soil, and biodiversity remain more important constraints than the number of floors in a farm.

Water is globally abundant and locally scarce. Urban water can be desalinated and recycled at a manageable energy cost in an expanded electric economy. The water required for food is of a different order of magnitude and must come mainly from rainfall, healthy soils, protected watersheds, and efficient irrigation. The conclusion is asymmetric: we can guarantee drinking water almost anywhere, but we cannot turn every desert into a vast agricultural region without very high ecological and energy costs. Post-scarcity requires adapting production to geography, not abolishing geography.

Housing is a mixed case. Materials and construction labor can become much cheaper, while well-located land, urban infrastructure, and permits can keep prices high. An economy with robots capable of erecting buildings quickly can still have a housing crisis if cities block supply or land rents absorb the productivity gain. In many domains, the final problem will not be manufacturing the object but the right to place it and access to the surrounding system.

1.2 The Scarcity That Remains and the Realistic Timeline

The first hard scarcity is lived time. Automation can free hours, and medicine can extend healthy life, but every human remains a sequential being. No one can read every book, live in every city, pursue every career, or sustain an unlimited number of deep relationships. As AI produces more options, the value of filtering and choosing rises. An abundance of answers creates a scarcity of reasons to trust any particular answer.

The second category consists of ecological and geographic goods. A mature forest, a reef, a species, a stable climate, or a historic city center is not merely an aggregate of matter. Such things can be protected, partly restored, or imitated, but they have a history and a location that cannot be copied without loss. Cheap energy can make more places comfortable and reduce pressure on nature; it can also accelerate extraction, transport, and destruction. The biosphere is not merely a supplier of inputs but a system with thresholds and irreversibilities.

The third category is relational. Trust, intimacy, love, admiration, and consent cannot be distributed like electricity. Other people's bodies and relationships are not resources to which material equality creates a right of access. A richer society can remove economic coercion from many relationships and make consent more genuine; it cannot guarantee everyone the partner, attention, or recognition they desire. Status competition may become more visible precisely here, because money can no longer purchase other people's dependence as easily.

Power is scarce in an even more direct sense. The capacity to decide for others is relational: not everyone can control the same institution. Technology can distribute tools, but infrastructures of energy, chips, models, and robots can create unprecedented centers of power. A materially abundant world can be democratic or authoritarian. Comfort is not the same as freedom, and free services can be offered in exchange for surveillance and conformity.

Status and positional goods will almost certainly survive. Not everyone can be in the top one percent, at the exclusive table, or the author of the first discovery. People compare not only whether they have enough but also their position relative to others. Realistic politics cannot eliminate every difference in prestige; it can prevent prestige from becoming a right to dominate, a monopoly over institutions, or control over other people's material lives. This is the difference between a society with functional hierarchies and one based on subordination.

There is no single date on which the world "enters" post-scarcity. Digital intelligence becomes cheaper faster than physical infrastructure; new energy capacity is built faster than cities; rich economies may universalize services decades before fragile regions. The transition will be a sequence of thresholds: first an abundance of translation, programming, design, and analysis; then automation of factories, warehouses, and transport; later general robotics in open environments; and latest of all, the reorganization of ownership and public services.

The book's central scenario places the possibility of a basic post-scarcity threshold in some rich economies between 2040 and 2060. A global version becomes plausible between 2070 and 2100. For 2050, I subjectively estimate a 35–55% probability that several high-income states will reach the threshold and a 10–20% probability that it will be reached worldwide. For 2070, the ranges rise to 65–85% in rich economies and 35–60% globally. By 2100, worldwide attainment is physically plausible but remains conditional on avoiding systemic wars, ecological collapses, and extreme concentration of infrastructure.

These percentages are not the output of a predictive model deserving decimal precision. They are prospective judgments intended to make the assumptions falsifiable. If electricity grids, mining,

robotics, and housing accelerate faster than I assume here, the timeline should be shortened. If robotic safety stagnates, geopolitical tensions fragment trade, or productivity gains are captured by rents, the timeline should be extended. The book offers a map to update, not an oracle.

The personal robot will probably emerge in stages and will resemble the automobile economically more than the phone. The first useful systems will be expensive, imperfect, and dependent on servicing, remote supervision, or simplified environments. Access through fleets, subscriptions, buildings, care services, and rentals will be more widespread than individual ownership. In the baseline scenario, fewer than 1% of the world's households have a dedicated general-purpose robot in 2030; 1–4% in 2035; 8–20% in 2045; and 25–50% in 2055. In rich economies, access could reach 5–15% in 2035, 25–55% in 2045, and 60–85% in 2055.

The decisive economic criterion is not the sticker price but the annual cost per hour of safe, useful work. A cheap robot that breaks down, requires a human operator, and cannot work around children is expensive. A more costly robot that operates for thousands of hours, lowers the cost of care, and can be upgraded is affordable as a service. Mass diffusion will begin when reliability, maintenance, and legal liability mature, not when the first spectacular demo appears.

1.3 The Final Obstacle: Institutions and Human Nature

Humanity does not appear programmed for a single social form. We are capable of extensive cooperation, sharing, reciprocity, and sanctioning dominators, but also of status competition, coalition-building, exploitation, and obedience. Hunter-gatherer societies described as egalitarian were not devoid of differences in age, gender, reputation, skill, or reproductive success. Their egalitarianism was often an active practice: mobility, compulsory sharing, consensus decision-making, ridicule of braggarts, and the ability to leave a leader. Agriculture, storage, inheritable land, large domesticated animals, and warfare made wealth more accumulable and domination easier to institutionalize (Borgerhoff Mulder et al., 2009; Kohler et al., 2017; Stibbard-Hawkes and von Rueden, 2025).

This history partly supports the observation that many people would rather exchange places with the dominator than abolish domination. Aversion to inequality is often stronger when a person loses than when that person gains. Stable systems offer those at the bottom the hope of mobility, identification with the dominant group, moral explanations of hierarchy, and small positions from which they can dominate someone even more vulnerable. The literature on system justification sometimes finds implicit preferences for advantaged groups among members of disadvantaged groups, but explicit findings are far more heterogeneous and include solidarity, resistance, and in-group preference (Essien et al., 2021). There is no law saying that the oppressed love oppression; there are simply enough mechanisms through which they can reproduce it.

Genetic variation is real, including for traits related to personality and orientation toward hierarchy, but it does not justify fatalism. Twin studies measure variation within a population in a particular environment; they do not discover a “gene for exploitation” and do not show that

institutions are irrelevant. A population may contain people strongly oriented toward dominance, people oriented toward prestige and contribution, conflict-averse people, and many intermediate combinations. Good institutions do not assume that everyone becomes moral; they limit the capacity of any temperament to convert local advantage into hereditary and uncontrollable power.

The disappearance of economic coercion does not imply the disappearance of motivation. Research on autonomous motivation shows that people pursue competence, autonomy, and relatedness, not only monetary reward. Guaranteed-income experiments have produced modest reductions in working time, not a general withdrawal from activity; in the U.S. experiment analyzed by Vivalt and colleagues, the reduction was roughly one to two hours per week (Vivalt et al., 2024). At the same time, involuntary unemployment harms mental health, while chronic boredom is associated with aggression and the search for destructive stimulation. An abundant society must provide structure, challenges, roles, belonging, and paths to prestige that do not depend on the threat of hunger.

The political verdict is harsher than the physical one. We can produce enough without providing enough. We can automate labor and concentrate ownership. We can guarantee goods while conditioning freedom. The most robust architecture would combine a universal floor of services, distributed ownership of productive infrastructure, genuine competition, limits on the conversion of wealth into political power, protection of privacy, and multiple paths to prestige through creation, knowledge, care, courage, and public service. Post-scarcity need not abolish ambition. It must make ambition compatible with the freedom of others.

This distinction also changes how the timeline should be read. There will be no morning on which the world officially wakes up “post-scarcity.” Some services will cross the threshold earlier: information, basic education, and part of diagnosis may become nearly free before housing, care, and mobility do. Some regions will gain abundant energy but remain constrained by water, administration, or conflict. Others will have strong distributive institutions but aging infrastructure. The transition will look like a map of islands of sufficiency, corridors of access, and abandoned zones, not a clean global line.

The human obstacle is not only the greed of an elite. It is also the collective difficulty of accepting that essential goods no longer need to be used as instruments of discipline. Some people will believe that security without compulsory work weakens character; others will defend privileges as evidence of merit; a minority will lose not comfort but the power to purchase obedience. The book does not assume these reactions will disappear. Its verdict is more modest: competitive and hierarchical predispositions can be channeled above a universal material floor, and institutions can prevent success from becoming an unlimited right over other people's lives.

There is also a cultural difficulty: modern societies have tied virtue to imposed effort and personal worth to the scarcity of one's position. If basic goods become easy to access, some old signals of merit will lose value. The reaction may be creative—new forms of excellence may emerge—or defensive, through the invention of new barriers, credentials, licenses, exclusivities, and ranks. The transition toward sufficiency therefore requires not only cheaper production but also a

culture capable of granting prestige for contribution, competence, care, and discovery without needing other people's poverty as a backdrop.

2. Abundance Is Not Infinity

2.1 What Sufficiency Means in Practice and How to Measure It

The post-scarcity fantasy is often expressed through an image: autonomous factories, nearly free energy, robots performing physical work, and artificial-intelligence systems designing, coordinating, and repairing everything. In this image, prices tend toward zero, work becomes optional, and distributive conflict loses its object. The image contains a valid intuition and a profound error. The intuition is that many goods are expensive not because their matter is scarce but because energy, information, coordination, and labor are required to transform it. If those costs fall by orders of magnitude, the material basis of life can become very cheap. The error is assuming that everything people want can be multiplied in the same way.

A safe, well-insulated apartment can be multiplied. A direct view of Lake Zurich cannot be multiplied without creating another lake or falsifying the experience. A standard medical consultation can be partly automated; the time of a particular doctor, the presence of a friend, and trust in a person do not scale in the same way. Music can be copied at almost zero cost; a front-row seat, an original work of art, and the status of being first remain scarce. Fred Hirsch called such things “positional goods”: their value depends on relative position or on a nearly fixed supply, not only on material utility (Hirsch, 1976). Technology does not eliminate competition; it often relocates it.

Post-scarcity must therefore be decomposed. A society can have an abundance of calories yet a shortage of healthy food in poor neighborhoods. It can have enough annual electricity-generation capacity yet suffer outages at peak hours. It can have millions of empty homes and homeless people. It can produce more information than it can evaluate and more options than people can choose among. Herbert Simon observed that an abundance of information creates a poverty of attention; in the age of generative models, this relationship becomes infrastructural (Simon, 1971).

In this book, post-scarcity does not mean the absence of all limits. It means that basic material limits are no longer used as the principal mechanism for disciplining the population. No one starves because their income disappears. No one freezes because energy prices rise. No one is excluded from elementary education or diagnosis because the supply of human labor is insufficient. Such a society may still have markets, income differences, luxury, competition, priority rules, and scarce goods. The difference is that survival and civic participation no longer depend on winning that competition.

To say that a good “exists in abundance” can hide five different questions. The first concerns the physical stock or flow. The second concerns transformation capacity. The third concerns marginal cost and price. The fourth concerns universal access. The fifth concerns resilience and sustainability. A country can pass the first three stages and fail at the last two.

Table 1. The Operational Scale of Abundance

Level	Decisive question	Example of failure
Physical	Is there enough natural flow or stock?	Solar energy is vast, but it does not arrive without installations.
Industrial	Can the resource be extracted and transformed at scale?	Copper exists geologically, but mines and refining expand slowly.
Economic	Does the full cost become low and stable?	An inexpensive solar park can remain inaccessible when financing is expensive.
Universal	Does the service reach almost every person?	Food is sufficient in aggregate, yet hundreds of millions suffer from hunger.
Resilient	Can the system withstand shocks without destroying the biosphere?	Cheap water from a depleted aquifer is not sustainable abundance.

Source: author’s synthesis and operational definitions.

This scale explains why purely technological arguments and purely political arguments are each insufficient. Without technology, we redistribute a limited and vulnerable quantity. Without institutions, productivity can feed luxury, monopoly, and waste before closing elementary deprivation. Streeck and colleagues estimate that closing global gaps in decent living standards would require roughly 12% more than the existing global stock of social material in 2015, if new construction were directed strictly toward those needs. At historical building rates, this volume could have been completed by 2030. But if countries follow the unequal accumulation pattern of rich economies, the additional requirement may increase nearly tenfold (Streeck et al., 2025). The material required for dignity is not exorbitant; the development model can become exorbitant.

Absolute scarcity appears when a stock cannot be replaced or regenerated at the relevant scale. Local scarcity appears when something is globally abundant but absent where it is needed: water in a desert, energy in an isolated grid, physicians in a rural region. Temporal scarcity appears when supply and demand do not coincide: solar electricity at noon, heating at night, medicines during an epidemic. Industrial scarcity appears when the material exists but there are not enough mines, factories, skilled workers, or transport. Ecological scarcity concerns the biosphere’s capacity to absorb disturbance: carbon, nitrogen, persistent substances, habitat loss. Institutional scarcity is created by rights, licenses, monopolies, war, corruption, or incapable administrative systems. Positional scarcity concerns goods whose value depends on exclusivity. Relational scarcity concerns trust, care, and authentic recognition.

Table 2. Types of Scarcity That Must Be Kept Distinct

Type	Mechanism	Can it be reduced?
Absolute	The quantity cannot be multiplied: time, a unique object.	No; only allocated.
Local	The resource exists, but not where demand is located.	Yes, through transport, networks, or migration.
Temporal	Supply and demand do not coincide in time.	Yes, through storage and flexibility.
Industrial	Extraction, refining, or production capacity is insufficient.	Yes, but often over decades.
Ecological	Regeneration or absorption capacities are exceeded.	Only through reduction, substitution, and circularity.
Institutional	Rights, prices, or power block access.	Yes, politically; technology alone is insufficient.
Positional	Value depends on others not having the same good.	Not at the level of the good itself; it can be reduced socially.
Relational	Trust, care, and legitimacy require history and reciprocity.	Partly; they cannot be manufactured at zero marginal cost.

Source: author’s classification, inspired by the economics of positional goods, commons theory, and the economics of attention.

These forms of scarcity can convert into one another. Cheaper energy can reduce the local scarcity of water through desalination, but it can increase ecological pressure if it produces more brine and urban expansion. Artificial intelligence can reduce the scarcity of expertise while intensifying the scarcity of attention and trust. Robots can reduce the scarcity of care labor without automatically reducing the scarcity of relationship: an elderly person may receive medication on time and still remain alone. A mature analysis asks not only which limit disappears but where the cost moves.

To avoid utopianism, I use a minimal criterion. A domain can be described as having reached basic post-scarcity only if the essential service reaches almost everyone, its cost does not exclude poor households, the system withstands foreseeable shocks, and provision does not transfer costs incompatible with climate stability, biodiversity, or public health. The threshold does not require identical outcomes, but a safe and robust minimum.

A basic service becomes universal only when energy, materials, food, water, robotics, distribution, and governance are all sufficiently robust. A single indispensable component that lags behind can block the outcome even if all the others have become abundant.

2.2 The Starting Verdict: Universal Sufficiency, Not Infinite Consumption

At the level of raw resources, humanity appears capable of supporting a population of roughly ten billion at a decent material standard. The UN projects a population peak of around 10.3 billion in the 2080s and a slight decline to 10.2 billion by 2100 (UN DESA, 2024). We therefore do not need to plan for a population that doubles indefinitely. Research on “energy for decent living” estimates that high universal standards could be supplied with roughly 125–183 EJ of final energy per year, depending on technology, urbanization, and population; inequality can raise the requirement to 269 EJ or more (Millward-Hopkins et al., 2020; 2022). In other words, civilization is not physically forced to choose between universal dignity and ecological limits. It is forced to choose among models of organization.

This conclusion must be held together with the opposite warning. Global resource extraction has risen from roughly 30 to 106 billion tonnes per year since 1970 and, in a scenario without major changes, could rise by another 60% by 2060. Resource extraction and processing account for a major share of emissions, pollution, and biodiversity loss, while rich countries use roughly six times more resources and generate ten times more climate impacts than low-income countries (UNEP, 2024b). We cannot extend the current material consumption of the richest to ten billion people. We can, however, extend the services that make life good while using less material per service, longer lifetimes, shared infrastructure, recycling, and clean energy.

The question, then, is not whether there is enough to reproduce the material lifestyle of the top percentile for everyone. Probably not, at least not without radical technological transformations and severe ecological costs. The question is whether there is enough for a civilization in which every person has what they need and very many people have much more. The preliminary answer is yes. The rest of the book explains why that “yes” is neither simple nor inevitable.

A universal threshold must be evaluated as a bundle of services, not as a national average. A country can produce enough electricity annually and still have disconnected households; it can have abundant calories and unhealthy diets; it can have housing units and homeless people. The test is whether the real person receives the service when needed, safely, and without a financial burden that pushes them back below the threshold. Resilience matters as much as price: a single fragile grid or a single monopolistic provider does not constitute abundance.

There is another essential boundary. The objective is not to multiply the material consumption of the richest percentiles across ten billion people. Much of that consumption is positional, wasteful, or dependent on externalizing costs. The objective is to multiply the services that sustain a good life: thermal comfort, healthy food, access, time, safety, and the possibility of participation. The same well-being can be produced through efficient buildings, excellent shared transport, and durable objects with far less matter than through individual ostentation.

The sufficiency threshold is inevitably historical. A decent home in 1900, one in 2026, and one in a society with autonomous robots do not contain the same bundle of capabilities. As general

infrastructure improves, what is necessary for social participation changes: connectivity, climate protection, access to cognitive tools, and digital security may become as important as running water. Universality does not require freezing a meager bundle in time; it requires updating it relative to technological conditions and to what is necessary to avoid exclusion.

At the same time, sufficiency needs an ecological ceiling. The fact that every person can receive energy, food, housing, and mobility does not imply everyone's right to any quantity of any good. A society can guarantee thermal comfort and transport without guaranteeing unlimited flights; it can guarantee habitable space without promising every household land in a historic center; it can guarantee a nutritious diet without turning every ecosystem into farmland. The social floor and biophysical limits must be designed together, or abundance for the present becomes scarcity for the future.

Above this threshold, markets and differences can continue, but without the right to block others' access to the necessities that allow them to refuse, leave, and participate.

3. Energy: The Nearly Universal Multiplier

3.1 From Solar Flow to the Infrastructure of Abundant Energy

Almost every form of material abundance is an energy conversion. Salt water becomes drinking water through pressure and separation. Low-grade ore becomes metal through crushing, heating, chemical reactions, and transport. Atmospheric nitrogen becomes fertilizer through energy-intensive processes. Mixed waste becomes raw material if it can be collected, sorted, and refined. A comfortable indoor climate is energy applied to a building. Computation is electricity transformed into inference. A robot is energy transformed into controlled motion.

For this reason, cheap energy does not guarantee post-scarcity, but its absence makes post-scarcity impossible. It lowers the cost of several constraints at once and enables substitutions: we can save land through more controlled agriculture, water through recirculation, high-grade ore through more intensive processing, and fossil fuels through electrification. Yet energy cannot substitute for everything. It does not instantly recreate mature soil, an extinct species, a stable climate, or a community of trust. It is a multiplier, not a magic wand.

Global energy demand grew by 1.3% in 2025, roughly 8 EJ, while electricity grew faster than total energy. Worldwide electricity consumption was estimated at 28,600 TWh in 2025 and is projected at 30,700 TWh in 2027 (IEA, 2026a; 2026b). Depending on accounting convention, the world energy system has an average flow on the order of 19–20 TW. This figure includes very large losses in heat engines, power plants, and industrial processes. An electrified economy can provide the same services with less primary energy because electric motors, heat pumps, and direct electric processes are more efficient.

For the projected peak population of 10.3 billion, a civilization operating at 20 TW would average 1.94 kW of energy power per person; 50 TW would mean 4.85 kW; 100 TW, 9.71 kW. These are not household allocations but totals for industry, transport, buildings, agriculture, computation, and public services.

Table 3. Orders of Magnitude for a Civilization of 10.3 Billion People

Average global power	Power per person	Indicative interpretation
20 TW	1.94 kW	Roughly the scale of the current system, with large fossil losses and highly unequal access.
30 TW	2.91 kW	Early electrified transition; allows expansion of essential services.
50 TW	4.85 kW	Lower threshold of the broad material-abundance scenario.
80 TW	7.77 kW	Automation, desalination, and clean industry at extended scale.
100 TW	9.71 kW	Energy-abundant civilization; ecological constraints become dominant.
150 TW	14.56 kW	Upper bound of the fast scenario; requires rigorous control of land use and heat.

Source: author’s calculations; peak population according to UN DESA, World Population Prospects 2024.

Models of “decent living” are instructive precisely because they do not require 100 TW. Millward-Hopkins and colleagues estimated roughly 15.3 GJ of final energy per person per year in a highly efficient and egalitarian 2050 scenario, equivalent to an average power of about 0.49 kW per person. Other context-sensitive estimates are higher, around 18–26 GJ. These levels do not describe the current energy luxury of the rich, but they include housing, mobility, food, health, education, communication, and infrastructure. They show that the material threshold of dignity lies far below consumption at the top of the distribution (Millward-Hopkins et al., 2020; 2022; Vogel et al., 2021).

Total solar radiation at Earth’s distance from the Sun is approximately 1,361.6 W/m² on a surface perpendicular to the rays. Multiplied by the planet’s cross-sectional area, this yields nearly 174,000 TW of incident power; averaged over the entire surface of the globe, it is about 340 W/m² (NASA, 2026). Capturing 0.01% of the incident flow would correspond to about 17.4 TW, 0.03% to 52 TW, and 0.1% to 174 TW. These tiny fractions do not mean panels can be placed anywhere. Clouds, latitude, night, efficiency, distance from demand, biodiversity, and competing land uses matter. But they dispel the idea that sunlight itself is a near-term limit.

The World Bank estimates that 93% of the world’s population lives in countries with average daily photovoltaic potential between 3 and 5 kWh per kWp, while the difference between the best and worst national averages is less than a factor of two (World Bank, 2020). The resource is more evenly distributed than oil or gas. Future energy inequality will depend more on capital, grids, land, institutions, and technology than on geology.

An order-of-magnitude calculation is useful. If a photovoltaic installation supplies an average net 40 W per square meter of collection area—an approximate assumption combining module efficiency and capacity factor—producing 50 TW would require about 1.25 million km² of collectors, roughly 0.84% of the planet's land area. The gross project footprint would be larger, but panels can be distributed across rooftops, parking lots, infrastructure, arid zones, bodies of water, and dual-use land. The scale is industrially enormous, not geographically absurd.

Worldwide renewable capacity increased by 692 GW in 2025, reaching 5,149 GW, a 15.5% increase in a single year (IRENA, 2026a). The weighted average cost of electricity from new projects was roughly USD 44/MWh for photovoltaics and USD 33/MWh for onshore wind in 2025 (IRENA, 2026b). The IEA expected renewables to become the largest source of electricity in 2026 and photovoltaic generation to grow by roughly 600 TWh during 2026 after a similar jump in 2025 (IEA, 2026b).

These figures must be interpreted carefully. The levelized cost of a solar farm is not the cost of an electricity system available every second. A grid with large shares of solar and wind needs transmission, flexibility, storage across different durations, demand management, and dispatchable sources. Very low marginal costs can coexist with large infrastructure investment. Financing can also make the same technology cheap in one country and expensive in another. Energy abundance is a system, not a panel.

Growth rates nevertheless have a decisive property: if a technology grows by 15% per year, capacity doubles in roughly five years. That pace will not remain constant indefinitely, but it shows that the physical base can change within a generation. The problem is that grids, permitting, workforce training, transformers, and storage capacity may grow more slowly than modules.

3.2 The Energy Mix and the Real Limits of Abundance

An abundant economy has no reason to be energy-monotheistic. Nuclear fission offers dense, dispatchable power with a small land footprint, but large projects in many jurisdictions have problematic capital costs and construction times. Standardized reactors, life extensions, and perhaps modular reactors can play important roles. Advanced geothermal could become a firm source in more regions if drilling techniques transfer from the oil industry. Hydropower remains essential where geography permits, but new dams carry ecological and social costs. Solar and wind will probably dominate growth because of speed and cost, not necessarily every hour of operation.

Fusion should be treated as an option, not a premise. In April 2025, the National Ignition Facility produced 8.6 MJ of fusion energy after 2.08 MJ of laser energy reached the target, a target-level gain greater than four. This result is scientifically important, but it is neither net gain for the entire facility nor a power plant (LLNL, 2026). ITER, the international magnetic-confinement experiment, plans to reach full magnetic energy in 2036 and begin deuterium-tritium operations in 2039 (ITER Organization, 2025). A significant commercial fusion plant before 2040 is unlikely; a substantial contribution to the global system before 2050 remains uncertain.

This is actually good news for post-scarcity realism. We do not need to wait for fusion. Solar, wind, fission, hydropower, geothermal, storage, and grids can build a 30–80 TW economy. Fusion could expand the safety margin and simplify some constraints after mid-century, but it is not the single key to the door.

Data centers consumed roughly 415 TWh in 2024, 1.5% of global electricity. The IEA projects roughly 945–950 TWh in 2030, close to 3% of the total, and around 1,200 TWh in 2035 in the central scenario; accelerated scenarios can be considerably higher (IEA, 2025a; 2026d). Growth is rapid and can create severe local bottlenecks because data centers are concentrated. At global scale, however, they are not an energy black hole. Even a tripling remains a fraction of an expanding electricity system.

The subtler question is the social return on energy used for computation. A MWh used to train or run a system that optimizes grids, designs materials, reduces losses, and automates maintenance may save multiples of its own consumption. The same MWh used to generate unlimited quantities of unwanted content may create little value and greater scarcity of attention. There is no purely technical answer; the economic architecture of demand matters.

All energy used on Earth ultimately becomes heat. At today's scale, the direct effect of anthropogenic heat is small globally, although it can matter in cities. At 100 TW spread across Earth's surface, the result is about 0.196 W/m^2 . At 500 TW, nearly 0.98 W/m^2 ; at 1,000 TW, 1.96 W/m^2 . These values would become comparable to major climate forcings and would impose limits even if the energy were entirely carbon-free. A terrestrial civilization operating at hundreds or thousands of terawatts cannot ignore thermodynamics.

That limit, however, lies far above the threshold required for material abundance. A highly efficient and circular 50–100 TW civilization would have two to five times today's energy flow and could provide much larger services because of electrification. In my baseline scenario, global useful energy reaches 30–45 TW around 2050 and 50–80 TW around 2100. A rapid scenario could reach 80–150 TW by the end of the century; a slow one, 25–40 TW. Beyond roughly 100–300 TW, heat, land, materials, and ecological disruption become increasingly difficult to treat as side effects.

There is no credible evidence that the planet's energy flow would prevent a good material life for ten billion people. There is strong evidence that construction speed, grids, storage, financing, and ecological limits can delay the transition by decades. Energy abundance is not an event that occurs when the price of a panel falls. It is the outcome of a long industrial mobilization.

The most reasonable conclusion is that energy can cease to be the dominant constraint on basic needs in many regions between 2040 and 2060. Globally, the full transition is more likely between 2060 and 2100. If we fail, it will not be because the Sun sent too little energy.

The word “maximum” must be used carefully. The immediate technical limit is how much we can build, not how much energy exists. The long-term ecological limit includes occupied land, materials, water, habitat fragmentation, and waste heat. At tens of terawatts, direct global heating remains small compared with greenhouse-gas forcing; at hundreds or thousands of terawatts, it can

no longer be ignored. A mature civilization should treat efficiency and heat dissipation as objectives rather than pursue raw power as a sign of progress.

Basic post-scarcity does not require a thousand-terawatt civilization. It is enough for clean energy to become several times more abundant, for grids to expand, and for uses to be electrified and efficient. Fusion could widen the frontier after mid-century, but the conclusion that enough energy exists does not depend on it. That makes the verdict more robust: we do not need to wait for a single breakthrough before beginning.

4. We Are Not Running Out of Atoms; Industrial Systems Are the Constraint

4.1 Reserves, Industrial Capacity, and the Case of Copper

When a technology grows rapidly, public discussion immediately shifts toward a list of materials supposedly destined to run out: copper for grids, lithium for batteries, neodymium for magnets, silicon for panels, sand for concrete. Some warnings are necessary. No industrial system can multiply demand without encountering constraints. But the image of a planetary tank being linearly drained is misleading.

In geology, a reserve is the identified quantity that can be economically extracted under current conditions. A resource is broader and includes known or estimated material that is not yet profitable to extract or fully characterized. A rise in price, a new separation technique, better infrastructure, or a discovery can move material from resource to reserve. Conversely, an environmental rule, a conflict, or declining ore quality can reduce the economic reserve. This is why the ratio of “reserves divided by annual production” is not a countdown clock to the extinction of a mineral.

The useful questions are different. How quickly can extraction grow? How much energy and water does it require? How concentrated is refining? How easily can the material be substituted? How much of the product can be recycled, and after how many years does it return to the flow? What local damage is produced? Is there enough geographic diversity to withstand an embargo? In an accelerated transition, “scarcity” is often the absence of capacity to turn ore into product when it is needed.

Worldwide extraction of biomass, fossil fuels, metal ores, and non-metallic minerals has more than tripled over the past five decades, from roughly 30 to 106 billion tonnes per year. UNEP estimates that without decoupling policies, the volume could rise by another 60% by 2060 relative to 2020 (UNEP, 2024b). This is not only a depletion problem. According to the report’s methodology, extraction and processing contribute to more than 60% of warming-related emissions and around 40% of the health impacts of air pollution. A post-scarcity economy built by permanently doubling the mass extracted would undermine the very system that supports it.

This tension requires a distinction between abundance of services and abundance of objects. A comfortable building lasting one hundred years can provide more service with less material per year than a poor building demolished after thirty. A mobility system combining trains, bicycles, shared vehicles, and compact urbanism can provide more useful trips with fewer tonnes of automobiles. A repairable robot with standard modules can produce more hours of work than three generations of sealed devices. The mature objective is not to maximize tonnes processed but to maximize years of useful service per tonne.

The materials required for abundance can be grouped into four families. Bulk structural materials—stone, sand, cement, steel, wood, aluminum—build housing and infrastructure. Conductive and electrochemical materials—copper, aluminum, lithium, nickel, graphite—support grids, motors, and storage. Functional materials used in small quantities—rare earths, gallium, germanium, indium, tantalum—provide magnetic, optical, or semiconductor properties. Nutrients and biological materials—phosphorus, potassium, fixed nitrogen, fertile soil, biomass—sustain the food system.

Table 4. Essential Materials: Geological Sufficiency and the Dominant Bottleneck

Material	Sufficiency verdict	Real bottleneck
Iron and steel	Geologically very abundant.	Decarbonizing reduction, infrastructure, and high-quality recycling.
Aluminum	Abundant.	Electricity, refining, and the impacts of bauxite extraction.
Copper	Large resources; supply flow is tight.	New mines, declining ore grades, refining; possible deficit by 2035.
Lithium	Resources probably sufficient for the transition.	Project speed, local water, refining, and battery-chemistry choices.
Rare earth elements	Not geologically “rare” in the simple sense.	Difficult separation, waste, and geopolitical concentration of processing.
Phosphorus	Important stocks, but no biological substitute.	Regional concentration, dispersion, and weak recovery from waste.
Sand, cement, aggregates	Enormous mass available.	Local ecological cost, transport, and process emissions from cement.
Biomass and wood	Renewable only within limits.	Land, biodiversity, regeneration time, and competition with food.

Sources: USGS (2026), IEA (2024; 2026), UNEP (2022; 2024); the synthetic assessment is the author’s.

Iron, aluminum, and silicon are abundant in the crust. Yet transforming them at large scale is energy-intensive and polluting. Copper is geologically widespread, but new large deposits often have lower grades, are more remote, or face legitimate opposition. Lithium is not extraordinarily rare, but production is concentrated, deposit chemistry varies, and scaling refining and the required water can be slow. Rare earth elements are so called for historical and mineralogical reasons, not because all are rare in the crust; the difficulty lies in separating them from chemically similar ores and managing the wastes.

Phosphorus belongs to a different category. It has no biological substitute: plants, animals, and humans need it. We do not “consume” it atomically, but we disperse it from mines into fields, waters, and sediments. A circular civilization must recover phosphorus and nitrogen from manure, wastewater, and organic residues. In this case, post-scarcity depends less on an infinite mine than on designing the metabolic loop between city and agriculture.

Electrification uses copper in grids, motors, transformers, vehicles, and buildings. USGS estimates that humanity has historically produced roughly 700 million tonnes, that identified deposits contain around 2.1 billion additional tonnes, and that undiscovered resources could raise the relevant geological total to about 6.3 billion tonnes (USGS, 2026a). Divided simplistically by a population of 10.3 billion, the total would exceed 600 kg per person. Not all of this material will be extracted, and the division says nothing about cost, but it shows that copper atoms are not disappearing in the next few decades.

The real bottleneck is flow. The IEA estimates that announced projects could leave a gap of roughly 25% relative to projected demand in 2035. Investment in critical minerals declined in 2025, while development lead times, ore quality, and concentration in refining amplify the risk (IEA, 2026c). A flow shortage can raise prices, delay grids, and change technologies even when the total resource is large.

The solutions combine. Aluminum can replace copper in some cables, with trade-offs in volume and connections. Motors can reduce or eliminate certain materials. Grids can be designed more efficiently. Copper recycling makes economic and energy sense, and the urban stock becomes an increasingly important “mine.” But material installed in a building or grid does not return for recycling for decades. During the expansion phase, recycling complements mining; once the installed stock matures, it can replace mining to a much greater extent.

4.2 Substitution, Circularity, and the Material Frontiers

USGS estimated worldwide lithium reserves in 2026 on the order of 37 million tonnes and resources of roughly 115 million tonnes (USGS, 2026b). These numbers must be read alongside technological dynamics. The desired function is electrochemical storage, not lithium consumption as an end in itself. Sodium-ion batteries, chemistries with less nickel and cobalt, iron-based stationary batteries, thermal storage, pumped hydro, compressed air, and demand management compete for different parts of the market. Relative scarcity changes design.

This is a general mechanism of technological abundance: functions are more substitutable than raw materials. We do not need every car to contain the same battery, every turbine to use the same magnet, or every building to use the same cement. Standardization helps cost, but technological monoculture can create artificial bottlenecks. A robust system preserves multiple functional pathways.

There are, however, limits to substitution optimism. An alternative technology has its own materials, factories, patents, safety tests, and learning curves. It cannot be activated instantly when prices rise. Diversity must be built before the crisis, which can appear inefficient in short-term accounting. Resilient abundance includes apparently “unused” capacity: inventories, multiple suppliers, alternative designs, and personnel who know how to operate them.

In public debate, the map of mines dominates. In practice, separation, purification, and intermediate processing can be more concentrated than extraction. The IEA reported that for a group of energy minerals, the average share of the largest refining country had reached roughly 72% in 2025, even excluding the most concentrated rare-earth chains. China holds the overwhelming majority of pretreatment and recovery capacity for batteries and dominant positions in graphite, magnets, and several specialty metals (IEA, 2026c).

This concentration is not merely a result of geology. It reflects decades of industrial policy, infrastructure, tolerance of environmental costs, vertical integration, and learning. Building a mine without a refinery may move rock, but it does not create industrial autonomy. For a post-scarcity economy, material security requires a geographically diverse network of refineries and standards that enable recycling. Otherwise, goods can be cheap on average and unavailable precisely during a conflict.

Critical materials are critical to a function within a technological architecture, not to civilization in the abstract. A battery can change chemistry, a motor can reduce permanent magnets, cables can combine copper and aluminum, and software can make better use of existing infrastructure. Substitution is not free: it can shift pressure onto another material, reduce performance, or require redesign. But it prevents a demand forecast for one technology from becoming a prophecy of collapse.

Refining deserves more political attention than it receives. Ore extracted in one country may depend on processing concentrated in another region, and a single trade incident can block many industries. Diversification, strategic stocks, agreements with social and environmental standards, recycling, and regional processing capacity are forms of insurance. They cost more than a supply chain optimized to the limit, but they purchase autonomy and continuity.

The largest material flows are not exotic metals but sand, gravel, stone, and cement. UNEP estimates global use of sand and aggregates at roughly 50 billion tonnes per year. Not all sand is suitable for construction; desert sand, rounded by wind, does not automatically replace aggregates with the required geometry. Extraction from rivers and marine areas can destroy habitats, alter sediment flows, and destabilize coasts. Here scarcity is local and ecological, not planetary.

Cement produces emissions both from fuel and from the chemical reaction that transforms limestone. Electrification does not eliminate the second source. Lower clinker content, supplementary cementitious materials, efficient structural design, component reuse, and carbon capture will need to be combined. Research published in 2026 argues that low-emission concretes could be scaled to meet global urban housing needs by 2050, but success depends on regional material availability and standards (Hafez et al., 2026). The conclusion is not that concrete becomes ecological through a single recipe, but that basic infrastructure is not blocked by a shortage of mineral mass.

Universal decent housing does not require every household to own a large villa. It requires adequate space, safety, privacy, thermal comfort, and access to services. Compact urbanism and durable buildings reduce material, energy, and travel time simultaneously. By contrast, suburban sprawl, premature demolition, and housing used as a speculative asset can consume more material without closing the housing deficit. Again, the problem is not only how much we can build but what we optimize for.

“Circularity” is sometimes presented as if all materials could circulate perfectly. Thermodynamics, wear, and contamination make a 100% loop impossible. Some materials disperse; alloys mix; fibers shorten; small products are lost; collection consumes energy. Real circularity has a hierarchy: avoid unnecessary material, extend lifetime, repair, reuse, remanufacture, recover components, and only then recycle at the material level.

The IEA estimates that expanded recycling policies and facilities could reduce the need for new mines by 2050 by roughly 40% for copper and cobalt and 25% for lithium and nickel in a scenario consistent with announced climate commitments (IEA, 2024a). Current rates for many energy minerals are around 10%, projected to approach 20% by 2040 under current policies. The potential is large, but not instantaneous.

Product design is decisive. If batteries, robots, and electronics are glued shut, encrypted, software-locked, or replaced every three years, recycling becomes an expensive operation for recovering scraps. If modules are standardized, material passports track composition, and the producer is responsible for end of life, the technical stock can circulate. The right to repair and interoperability are resource policies, not merely consumer rights.

Asteroid mining is often invoked as the ultimate solution for metals. Physically, the Solar System contains resources incomparably larger than the terrestrial economy. Economically, the problems are moving mass, autonomous operations, uncertainty about composition, initial investment, and the price collapse that would follow a large increase in supply. In the first half of the century, space resources are more likely to be used in space—water for fuel and life support, regolith for construction—than transported to Earth in bulk.

Asteroids are not necessary for the basic abundance threshold. Earth’s crust, recycling, and substitution provide enough matter if we accept efficient design and internalize ecological costs. After 2050 or 2070, an autonomous space industry could relax limits on orbital infrastructure, space-based solar power, or off-planet computation. It extends civilization’s possible maximum; it does not automatically solve poverty on Earth.

There is no single “resource crisis.” There is a sequence of bottlenecks: design, permitting, extraction, refining, manufacturing, transport, recycling, and distribution. Over the next two decades, copper, certain battery materials, transformers, low-emission cement, and refining capacity can slow the transition. After mid-century, if the installed stock becomes large and products are recyclable, pressure on mining can decline in relative terms.

My estimate is that humanity can build the material mass needed for a decent global threshold by 2050–2070. Studies suggest that the strictly necessary additional mass is surprisingly small compared with the existing stock. But simultaneously building decent infrastructure and extreme material consumption for elites can double or triple the requirement, destroying ecological feasibility. Matter does not force the world to be poor. Inefficiency, positional luxury, and distribution can nevertheless make the material world impossible to satisfy.

Asteroid mining could become relevant for highly valuable materials used in space or for orbital infrastructure, where avoiding launch from Earth changes the economics. As a solution for the cement, steel, and copper needed by terrestrial cities in the coming decades, it remains a distraction. Cost, risk, and development time are too high relative to improving mines, recycling, and design on Earth.

The material verdict is therefore conditional, not triumphalist. There is enough matter for a decent and probably comfortable standard of living for all. There is not enough ecological space to replicate unchanged the current material metabolism of elites. Abundance requires more service per tonne, repairable objects, durable buildings, and an economy in which growth in value does not automatically mean growth in extracted mass.

5. Food, Water, and the Living Planet

5.1 Food, Land, and the Problem of Access

Food is the moral test of any theory of resources. If a system produces enough in aggregate and people still go hungry, the problem can no longer be described only in tonnes. FAO estimates that 645 million people experienced hunger in 2025, 7.8% of the world's population. Roughly 2.1 billion experienced moderate or severe food insecurity, and 2.7 billion could not afford a healthy diet (FAO et al., 2026). These figures coexist with obesity, waste, and the use of crops for feed, fuel, or products with low nutritional value.

Contemporary hunger is produced by poverty, conflict, prices, loss of livelihoods, weak infrastructure, discrimination, and climate shocks. Drought can reduce local supply, but whether it becomes famine depends on markets, stocks, transport, political power, and households' ability to buy. Amartya Sen showed that legitimate access to food—"entitlements"—can fail even when food exists within a country (Sen, 1981). A post-scarcity economy must be measured by people's nutrition, not by record harvests.

OECD and FAO project roughly 13% growth in global agricultural and aquatic production between 2026 and 2035, mainly through productivity and intensification rather than proportional land expansion. At the same time, direct agricultural emissions would rise by roughly 6% in the baseline scenario (OECD-FAO, 2026). A scenario analysis in the previous edition of the report showed that eliminating undernourishment by 2034 while reducing direct agricultural emissions would require roughly 10% additional production and a 15% productivity increase relative to the reference path, using available technologies (OECD-FAO, 2025).

These values are ambitious, but they do not require multiplying production several times over. Population is stabilizing, and the greatest opportunity lies in reorganizing flows. In 2022, 1.05 billion tonnes of food were wasted at retail, food service, and household levels: 132 kg per person and nearly 19% of food available to consumers. Households accounted for 60% of the total, food service 28%, and retail 12% (UNEP, 2024a). Another 13.2–13.3% is lost between harvest and retail (FAO, 2024a; 2025a). The percentages cannot simply be added to say that one-third is recoverable because the denominators differ and some of the mass is inedible. Still, the scale shows that the system contains an efficiency reserve comparable to the growth required.

Not all calories use land in the same way. Animals can convert grass from land unsuitable for crops and can use residues, but much of the livestock system consumes crops and land that could feed people directly or return to nature. The global study by Poore and Nemecek, based on roughly 38,000 farms, documented enormous variation among producers and high average impacts for many animal products (Poore and Nemecek, 2018).

A scenario modeled by Kozicka and colleagues replaces 50% of pork, chicken, beef, and milk globally with plant-based alternatives by 2050. The result is a 31% reduction in agricultural and land-use emissions relative to 2020 and an almost complete halt to net loss of forests and natural lands; restoring part of the freed land would double the potential climate benefits of the land sector (Kozicka et al., 2023). The model does not automatically predict consumer preferences and does not capture every nutritional issue. It does show, however, that protein choice is land policy at continental scale.

An abundant economy need not become uniformly vegetarian. It could combine predominantly plant-based diets, animals raised in circular systems on pasture and by-products, precision fermentation, microbial proteins, low-impact aquaculture, and cultivated meat where competitive. Rarely does a single technology win everything. The objective is to separate culinary and nutritional satisfaction from disproportionate pressure on land.

Controlled-environment agriculture can produce perishable vegetables near cities with water recirculation, little pesticide use, and high yields per area. For plants whose valuable parts have low mass and high price—lettuce, herbs, seedlings, certain fruits—it can make sense. For wheat, maize, rice, or potatoes, artificial light converts electricity into photons and then into biomass with substantial losses. A field uses sunlight for free across a large area.

Vertical farming therefore does not “free all land.” It becomes a specialized component: local production, resilience, planting material, high-value crops, and regions with harsh climates. Solar greenhouses, genetic improvement, sensors, precision irrigation, agricultural robots, and forecasting may have broader impact. Food post-scarcity will probably be hybrid agriculture, not a universal food factory.

Agricultural land is not merely area. Fertility, water, climate, access, biodiversity, and proximity to infrastructure matter. A diet rich in animal products uses land multiple times: for grazing and for animal feed. Reducing the most land-intensive part of consumption can free areas for nature without reducing nutrition. This is one of the few interventions that can simultaneously improve food security, climate, and biodiversity, but the transition must respect food culture and avoid poor-quality ultra-processed substitutes.

Controlled agriculture will occupy valuable niches: seedlings, short-cycle plants, perishables, pharmaceuticals, and production in regions with extreme climates. For wheat, maize, rice, or potatoes, artificial lighting turns electricity into an expensive substitute for sunlight. Technology can reduce land use for certain products; it does not abolish the thermodynamic advantage of photosynthesis in the field.

5.2 Water, Soil, Climate, and the Limits of the Living Planet

Globally, agriculture accounts for roughly 69% of freshwater withdrawals, municipalities 12%, and industry 19%, with large regional variation (FAO AQUASTAT, 2026). Nearly half of the population experiences severe water scarcity during at least part of the year, and one quarter lives in countries under extreme stress, where withdrawals exceed 80% of available renewable resources (UNESCO, 2024). Climate change alters precipitation, snow, glaciers, and evapotranspiration, making infrastructure designed around the past less reliable.

Desalination is often either idealized or dismissed. Seawater reverse osmosis in modern plants consumes roughly 2.2–3.5 kWh/m³, close to current practical limits, excluding all intake and distribution costs. It also produces brine and uses chemicals that must be managed (Kim et al., 2021). At this energy intensity, domestic water for all humanity is not physically prohibitive; agricultural water is.

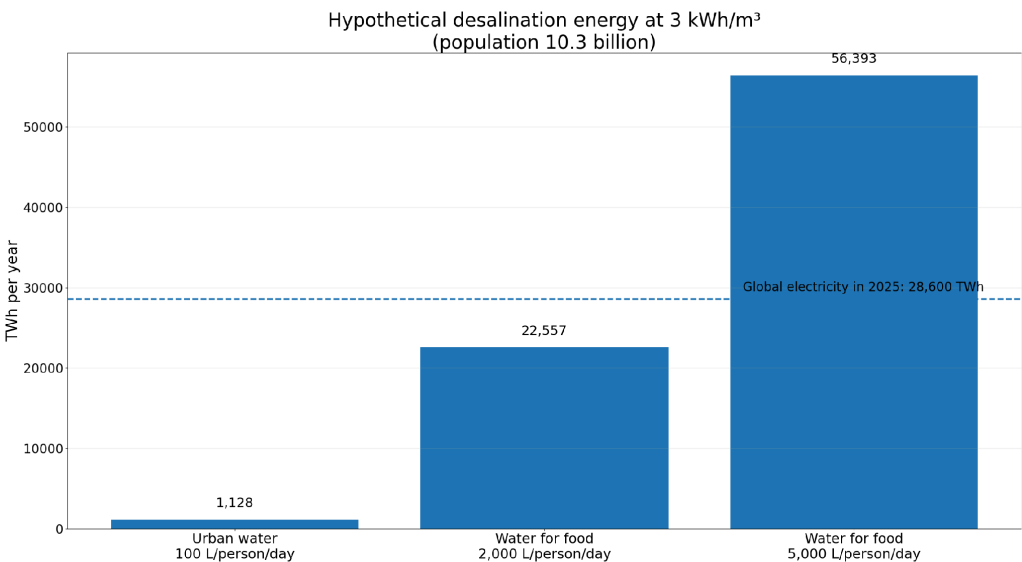


Figure 1. Annual Energy Required for Desalination at Peak-Population Scale

Source: author’s calculations from the water requirements discussed in the text; global electricity benchmark uses IEA, 2026.

For a population of 10.3 billion and 100 liters per person per day, annual volume is roughly 376 km³. At 3 kWh/m³, it would require about 1,128 TWh annually, roughly 4% of global electricity in 2025. In reality, we would not desalinate all water: rain, rivers, groundwater, and reuse remain cheaper. The calculation shows a safety net for coastal cities, not a universal strategy.

FAO estimates that a person's daily food embodies roughly 2,000–5,000 liters of water, depending on diet and conditions. If we hypothetically produced this water by desalination for 10.3 billion people, the energy requirement would be about 22,600–56,400 TWh per year, between 79% and 197% of global electricity in 2025, before pumping. Most of this water is actually “green” water—rainfall stored in soil—and does not need to be delivered through pipes. The conclusion is clear: with enough energy and capital we can provide drinking and urban water almost anywhere; we cannot treat the agricultural ecosystem as a planet-sized hydroponic installation supplied from the ocean.

Fertile soil is a living structure formed over time: minerals, organic matter, microbes, pores, water, and roots. It can be improved, but it is not an interchangeable industrial powder. Erosion, salinization, compaction, and carbon loss can reduce productivity even when fertilizer and energy are available. Synthetic nitrogen made feeding billions possible, but surpluses produce eutrophication, nitrous oxide emissions, and aquatic dead zones. Phosphorus lost to waterways contributes to the same problem.

The planetary-boundaries framework is imperfect and contains methodological uncertainty, but it offers a systemic warning. The 2025 assessment considers seven of nine boundaries transgressed: climate, biosphere integrity, land-system change, freshwater, biogeochemical flows, novel entities, and ocean acidification. Only stratospheric ozone and global aerosol loading are assessed as remaining in the safe zone (Planetary Health Check, 2025). “Transgression” does not mean the system collapses tomorrow; it means the risk of large and irreversible changes rises.

This is the fundamental limit of a purely industrial vision. We can produce nitrogen, water, light, and climate control, but we cannot cheaply replace all ecosystem services: pollination, stabilization of local climate, nutrient cycling, coastal protection, genetic diversity, and resilience to disease. A post-scarcity economy must leave space for nature not out of romanticism but because the biosphere is infrastructure of a complexity we do not know how to reconstruct.

The Global Carbon Budget 2025 estimated that fossil CO₂ emissions remained at record levels and that the remaining budget for a 50% chance of limiting warming to 1.5°C was on the order of 170 billion tonnes of CO₂—about four years of emissions at the 2025 level (Global Carbon Project, 2025). Climate budgets contain uncertainty and depend on the chosen probability, system response, and non-CO₂ emissions. The robust message is that the period during which we can build new fossil infrastructure without abandoning it early is very short.

Future abundance therefore has a sequencing problem. We must build energy, housing, grids, and industry for billions of people while cutting emissions rapidly before clean infrastructure is complete. If the transition uses a temporary boost from fossil fuels, the climate budget shrinks. If it is too slow, climate shocks destroy part of the gains. Speed is not merely an economic preference but a physical constraint.

Technically, humanity can provide enough calories, protein, and micronutrients by mid-century. Refrigeration, rural infrastructure, resilient varieties, efficient irrigation, loss reduction, diversified trade, stocks, and social protection are known technologies and institutions. No miraculous discovery is required to eliminate hunger.

It is harder to do this while preserving soil, water, climate, and biodiversity. A system can be post-scarcity for the consumer for several decades and ecologically bankrupt in the long run. This is why the book's criterion includes sustainability and resilience. Cheap food obtained by exhausting an aquifer is not abundance; it is transferring the bill to the future.

My estimate is that involuntary hunger could be eliminated globally by 2040–2050 with existing technology if there were relative peace, social protection, and logistical investment. Political probability is lower than technical feasibility. A universally affordable healthy diet and a food system compatible with ecological limits are more plausible between 2050 and 2070. Safe drinking water can approach universality in the same interval, but hydrological security for agriculture will remain regional and vulnerable.

Basic food may become abundant before agriculture becomes ecologically safe. Fertilizers, irrigation, and breeding can sustain production even as soils degrade and habitats disappear. That is a short-term success and a long-term debt. The post-scarcity threshold must include stability of the biological base, not just calories in the current year.

Hunger will most likely be eliminated through a combination of income, peace, stocks, logistics, and public programs before it is eliminated through radical food technologies. Precision fermentation, cultivated meat, and gene-edited crops can reduce pressure, but distribution remains decisive. Hunger is a test of institutions: if it persists in a world capable of producing enough, the primary cause is access.

6. When Thought Becomes Cheap, the Physical World Remains Slow

6.1 Cheap Intelligence in a Slow Physical World

Artificial intelligence and robotics are often presented as a single revolution. Economically, they follow different curves. A software model can be copied into millions of instances once it has been built. A robot must be manufactured, transported, powered, maintained, and insured. An algorithmic improvement can be distributed overnight; a better joint must enter an industrial supply chain and be physically replaced. Information scales with almost no mass. Action remains tied to mass, friction, and risk.

This asymmetry makes likely a period in which design, analysis, code, translation, advice, and coordination become extremely cheap while housing, physical care, construction, and repairs remain relatively expensive. We will be able to generate thousands of building designs in an hour, but not pour the foundations in that same hour. We can produce optimal plans for a grid, but transformers and lines must be built. Cognitive abundance may even increase demand for physical execution and make bottlenecks more visible.

Evidence from 2023–2026 shows significant gains, but concentrated in tasks. Brynjolfsson, Li, and Raymond studied the introduction of a generative assistant for 5,172 support agents and found an average increase of roughly 15% in problems resolved per hour, with much larger effects for less experienced workers (Brynjolfsson, Li, and Raymond, 2025). Noy and Zhang found that participants performing professional writing tasks finished faster and with higher average quality when using a generative model (Noy and Zhang, 2023). Other experiments in programming, consulting, and collaboration show gains ranging from a few percent to more than 25%, but also errors when the task exceeds the system's competence frontier.

These results do not automatically translate into macroeconomic productivity. Organizations must change processes, responsibilities, training, software, and incentives. A task saved may be replaced by verification, integration, or increased volume. OECD notes that worker-level evidence is stronger than evidence for aggregate gains and that skill and reorganization barriers are important (OECD, 2025a; 2026a). In 2025, more than one-third of people in the OECD had used generative tools, but adoption remained unequal (OECD, 2026b).

ILO estimates that roughly one job in four has some degree of exposure to generative AI, but task transformation and augmentation are more likely in the near term than complete automation of occupations (ILO, 2025). An occupation is a bundle of tasks, relationships, responsibility, and context. Even if 60% of a lawyer's text is generated, signature, strategy, negotiation, and liability may remain human. The cost of the service can fall without the profession disappearing.

An industrial environment is designed for robots: standardized objects, known positions, flat surfaces, separation from people, or controlled speeds. A home is the opposite. Clothes deform, drawers jam, animals move unpredictably, fragile objects are mixed with trash, and the same instruction—“tidy up”—depends on cultural and personal preferences. A chatbot error is a wrong sentence; a robot error can break a glass, injure a child, or start a fire.

General physical autonomy requires robust perception, fine manipulation, spatial memory, prediction of people, planning, fast closed-loop control, fault detection, and safe behavior in unfamiliar situations. Each function can be demonstrated separately; integrating them at a product cost and an acceptable error rate is the problem. A robot that succeeds in 95% of actions looks excellent in a video and intolerable if it performs a thousand actions per day: it would produce fifty failures.

For many domestic tasks, reliability must exceed 99.9%, and the remaining failures must be mild and detectable. This explains why teleoperation and human supervision are reasonable bridges. They make the product useful before autonomy is complete and generate training data. But they shift part of the cost back into labor and create problems of privacy, security, and jurisdiction.

The International Federation of Robotics reported 542,000 industrial robots installed in 2024 and an operational stock of 4.664 million, with 74% of new installations in Asia (IFR, 2025a). In professional services, sales were close to 200,000 units, while consumer robots reached nearly 20 million units in 2024, mainly vacuum cleaners and lawn mowers (IFR, 2025b). These are already forms of mass-produced autonomous labor.

The lesson is that autonomy spreads first when the environment or task is constrained. A robot vacuum has no arms and does not wash dishes, but it can eliminate hundreds of hours of work over its lifetime. An automatic washing machine, although not called a robot, is a manipulation and control system that changed domestic labor more than many humanoid demonstrations. Post-scarcity may arrive through an ecosystem of specialized machines before a universal anthropomorphic servant.

The humanoid form nevertheless has advantages. The built world is adapted to the human body: stairs, handles, tools, shelves, vehicles. A robot with similar dimensions and limbs can use existing infrastructure. The disadvantages are mechanical complexity, stability, and cost. The market will probably combine humanoids, fixed arms, mobile robots, exoskeletons, autonomous vehicles, and specialized appliances.

6.2 The Economics of Robotic Labor and the Threshold of Near-Free Work

In 2026, Unitree listed the R1 humanoid platform from USD 4,900, the G1 at USD 13,500, and the H2 at USD 29,900. 1X offered the NEO domestic robot for USD 20,000 for early access or USD 499 per month, with U.S. deliveries beginning in 2026. NEO was described as having basic autonomy and the option for a human expert to remotely supervise complex scheduled actions (Unitree, 2026; 1X, 2026). These prices mark an important shift: humanoid mechanics have entered the price range of an automobile, not a multimillion-dollar industrial installation.

But a USD 20,000 automobile is useful from the first day within a mature infrastructure. It has standards, service networks, parts, insurance, trained drivers, and a clear function. A robot at the same price may be a research platform, may require supervision, and may perform few useful tasks. The correct comparison is total annual cost divided by the hours of valuable work performed safely.

The annual cost of a robot includes depreciation and financing of the purchase price, maintenance, software, energy, and insurance. This full cost, divided by the hours in which the robot safely performs a useful task, is more relevant than the sticker price.

In an early scenario, a USD 20,000 robot used for seven years, with financing, high maintenance, software, energy, and insurance, would cost around USD 6,460 per year, roughly USD 540 per month. In a mature scenario, a USD 10,000 product with a longer life and standardized servicing would fall toward USD 2,760 per year. A mass-market USD 5,000 product operated for ten years could reach roughly USD 1,270 annually. These figures are transparent scenarios, not price forecasts.

Table 5. Total Annual Cost of a Personal Robot in Three Phases

Stage	Main assumptions	Annual cost and indicative income threshold
Early	USD 20,000; 7 years; 7% cost of capital; maintenance 8%; software USD 1,500/year.	≈ USD 6,461/year (USD 538/month); comfortably affordable at ≈ USD 65,000 income if the threshold is 10%.
Mature	USD 10,000; 8 years; 5%; maintenance 5%; software USD 600/year.	≈ USD 2,757/year (USD 230/month); income threshold ≈ USD 28,000.
Mass market	USD 5,000; 10 years; 4%; maintenance 3%; software USD 200/year.	≈ USD 1,266/year (USD 106/month); income threshold ≈ USD 12,700.

Source: author’s cost model. Excludes the value of time saved, local taxes, and subsidies.

If the early robot provides only two useful hours per week, the cost is prohibitive. If the mature version provides two hours per day, the cost can fall below USD 4 per hour. At six useful hours per day, it can transform the economics of care, cleaning, food preparation, and maintenance. The decisive progress, therefore, is not the price falling from USD 20,000 to USD 10,000, but useful autonomous hours rising from tens to thousands per year.

The price of housing includes land, regulation, financing, and location, not only construction labor. The price of a medicine can include patent monopoly and regulation. Food prices include land, climate risk, distribution, and market power. If robots reduce labor costs by 80%, the final good may fall by 20% or not at all if other rents absorb the gain.

This is an important lesson from earlier automation. Manufacturing productivity made electronics, textiles, and many goods very cheap, while housing in attractive cities, health care, and education became more expensive in many countries. The Baumol effect partly describes rising costs in slow-productivity sectors when wages must compete with more productive sectors (Baumol and Bowen, 1966; Baumol, 2012). AI and robots can weaken this cost disease in services, but land, liability, and human relationship remain.

The most likely transition is not “humans or robots” but a new division of labor. AI will plan, explain, simulate, and monitor. Robots will execute repetitive or dangerous physical tasks. Humans will define goals, manage exceptions, assume responsibility, and provide relationship where it matters. As systems improve, the frontier moves, but not all activities have the same value when fully automated.

In elder care, a robot can lift a person, bring objects, monitor falls, and prepare medication. This can free staff for conversation and judgment. In a system oriented exclusively toward cost reduction, the robot can instead be used to eliminate staff, leaving the person alone. The same technology produces relational abundance or relational poverty depending on the institution.

“Near-free labor” should mean a very low marginal cost for an additional hour after capital is amortized. For standardized cognitive tasks, this threshold is already visible: generating a translation, summary, or preliminary draft costs cents or less, although verification remains. For repetitive industrial tasks, robotic cost can be very low in well-designed factories. For general domestic work, construction in variable environments, and care, the threshold is farther away.

My central estimate is as follows. Between 2026 and 2035, the cost of many cognitive tasks falls by one to two orders of magnitude, but organizations absorb only part of the gain. Between 2030 and 2045, robots become common in logistics, manufacturing, agriculture, professional cleaning, and modularized construction. Between 2035 and 2055, general domestic robots move from experimental products to useful appliances in rich countries. Between 2045 and 2070, the cost of standard physical labor can become low enough that basic goods are constrained more by energy, materials, land, and institutions than by human labor hours.

This timeline can shift by ten or twenty years. Software progress may accelerate abruptly. A major safety incident, regulation, war, or chip bottlenecks can slow it. But the general direction is robust: information becomes cheap first, physical action follows, and institutions change slowest of all.

“Near-free” will not mean costless. Even when robots and AI perform the task, energy, capital, maintenance, supervision, insurance, safety, and liability remain. For rare or dangerous activities,

these costs may dominate. It is more accurate to say that human hours become a small component of cost for an expanding set of standardized goods.

The economic effect will be uneven. Products that can be designed, manufactured, and transported in controlled systems will become cheaper first. Services involving trust, vulnerable bodies, messy environments, or moral decisions will retain a human component longer. Paradoxically, authentic human labor may become a luxury in some domains precisely when mechanical labor becomes commonplace.

7. What Cannot Become Abundant

7.1 Time, Attention, Places, and the Biosphere

When the cost of one input falls, the economy reorganizes around the next constraint. The pattern is old. Printing reduced the scarcity of text copies and increased the importance of literacy, distribution, and selection. The internet reduced the scarcity of publishing and created a crisis of attention, authenticity, and trust. Artificial intelligence reduces the scarcity of producing answers and increases the scarcity of evaluating them. Robots will reduce the scarcity of execution and increase the importance of goals, access, safety, and legitimacy.

A world with cheap energy, intelligence, and labor is therefore not a world without economics. It is a world in which prices and conflicts concentrate around things that cannot be replicated through computation and actuation. Some of these are hard physical limits. Others are relational or institutional. A few are deliberately created to preserve profit or status.

Table 6. What Will Remain Scarce in an Economy with Cheap Intelligence and Labor

Scarce resource	Why it cannot simply be multiplied	Likely effect
Time and attention	Each person has a finite life; content supply grows faster than the capacity to experience it.	Selection and filtering become more valuable than production.
Desirable places	A historic center, shoreline, or community cannot be copied in the same location.	Land rents and conflicts over access.
Ecological capacity	Climate, biodiversity, soils, and oceans have regeneration and absorption thresholds.	Caps on material flows even when energy is cheap.
Safety and acceptable risk	Powerful systems can cause rare but catastrophic harms.	Verification, liability, and control become central costs.
Privacy	Copied data cannot be “unknown” again; cheap surveillance erodes autonomy.	Unobserved spaces acquire economic and political value.
Trust and authenticity	There is no complete shortcut for history, reputation, and reciprocity.	Proofs of origin and real relationships become premium goods.
Power and legitimacy	Collective decisions cannot be optimized without deciding who has the right to choose.	Conflict shifts from production to governance.
Status and uniqueness	Positional goods lose their function if they become universal.	New hierarchies emerge even after basic shortages are eliminated.

Source: author's conceptual synthesis; see Hirsch, Simon, and Ostrom.

Each person has roughly the same number of hours in a day. Medicine can extend healthy life, and automation can free time, but subjective experience remains sequential. We cannot read every book simultaneously, live every career, or maintain an unlimited number of close relationships. When the production of options explodes, the time required to choose and experience them becomes more valuable.

This changes the meaning of wealth. In the industrial economy, wealth means the capacity to buy objects and services. In a materially post-scarcity economy, wealth may mean control over one's own time, freedom from interruptions, health, mental space, and access to experiences that cannot be accelerated. A system that provides cheap goods but colonizes every minute through surveillance work, notifications, and reputational competition is not abundant in the human sense.

Generative models can produce more texts, images, films, games, and hypotheses than the entire population can consume. The marginal cost of content tends toward zero; the cost of deciding what is worth seeing rises. Attention becomes a rival resource: an hour given to one object cannot be given to another. Recommender systems, editors, reputation, and brands become allocation infrastructures.

This scarcity can be exploited. Platforms that control filtering can extract rents even when content is abundant. They can favor dependence, outrage, or conformity because these maximize captured time. In a world of nearly free cognitive production, epistemic institutions—verification, traceability, provenance, reputation, and critical evaluation—are as important as factories. We will not lack answers; we will lack good reasons to believe one of them.

The planet's surface cannot be multiplied, and certain places combine climate, safety, services, beauty, and social networks. Even if robots build housing almost for free, central land in Zurich, Geneva, Paris, Singapore, or a quiet coastal village remains limited. Technology can create new cities, faster transport, virtual reality, and microclimates. It can make more places good. It cannot make everyone live in the same historic center.

This scarcity explains why housing can remain expensive in an automated economy. The cost of the structure can fall while land rent absorbs the gain. If urban supply is blocked, landowners capture the robot's productivity. Housing policy, transport, and taxation of land rents are therefore components of post-scarcity, not later details.

Energy and matter can be partly recycled; ecosystems have thresholds, delays, and irreversibilities. A mature forest is not merely a quantity of carbon or timber. A reef is not a replaceable mineral structure. An extinct species does not return as GDP rises. Even if synthetic biology recreates some functions, the complexity of ecological relationships and historical evolution cannot be cheaply replicated.

The capacity of the atmosphere and oceans to absorb carbon without dangerous change is scarce. The capacity of rivers to absorb nitrogen and chemicals is scarce. Space for connected habitats is scarce. These are “common-pool resources” in Ostrom’s sense: resources for which exclusion is difficult and overuse by one actor reduces quality for others (Ostrom, 1990). Private abundance built on degradation of the commons is an accounting illusion.

Technology can create partial substitutes for the scarcity of place: rapid transport, telepresence, well-designed new neighborhoods, and controlled indoor climates. But substitution has limits. Proximity to family, history, and nature is not merely a high-resolution image. Land policy and urban planning will therefore decide how much of construction abundance reaches the citizen and how much becomes rent for the owner of the location.

For the biosphere, the prudent principle is not to use technical restoration capacity as permission to destroy. Ecosystems can have unknown thresholds, and rebuilding them can take centuries. A rich economy must purchase ecological redundancy: larger habitats, corridors, cleaner waters, and safety margins. Durable human abundance requires space that is not optimized for human use.

7.2 Safety, Trust, Power, and Positional Goods

In a factory, a robot can be stopped and the area isolated. In a home, an autonomous system interacts with children, sick people, fire, sharp objects, and intimate data. Safety cannot become free because it requires testing, redundancy, auditing, insurance, updates, and liability. As the number of autonomous systems grows, even very low failure rates can produce many incidents.

There is also systemic risk. If billions of robots, vehicles, and infrastructures use the same models, a software vulnerability or defective update can have simultaneous effects. Diversity, degraded modes, local control, and the possibility of manual operation carry costs, but they buy resilience. An abundant society needs more safety engineering, not less.

A useful domestic robot sees almost everything: cameras, voices, bodies, medicines, habits, conflicts, visitors, and objects. Teleoperation can send this world to a human operator. Models can infer health, relationships, and preferences without the user ever declaring them. Privacy becomes scarce when cheap services are financed through surveillance.

Technically, local processing, encryption, compartmentalization, and data control can better protect private space. Economically, providers have incentives to collect data for improvement, monetization, and liability reduction. The right to a robot that works without turning the home into a data stream for a monopoly may become the twenty-first-century equivalent of the right to private correspondence.

Safety is a good with rising marginal cost. Moving from 95% to 99.9% reliability may require more work than the first leap from prototype to product. In a home, rare errors are inevitably multiplied across billions of hours of operation. Standards, incident reporting, independent testing,

and clear liability will determine whether domestic robotics becomes infrastructure or a succession of tolerated accidents.

Privacy must be designed into the architecture, not offered through a privacy policy. Local processing, minimal data retention, offline modes, identity separation, and access auditing are real costs. A market that subsidizes the robot by monetizing life inside the home may make the device cheap and freedom expensive. Post-scarcity is not compatible with the idea that poor people receive automation only in exchange for surveillance.

Trust is not merely the prediction that an agent will do what it says. It includes shared history, reciprocal vulnerability, and moral accountability. A system can simulate empathy perfectly while having neither interests of its own nor the capacity to suffer. For some purposes, simulation is sufficient: a calm interface can reduce anxiety. For others, the distinction between relationship and service remains essential.

As voice, image, and style become imitable, evidence of provenance and verifiable encounter gain value. “Made by a human” may become a luxury category or moral signal, much like handcrafted objects after industrialization. But authenticity should not become a cult that rejects the benefits of automation. The question is where human origin is part of the product: a love letter, testimony, a vote, a judicial decision, an artwork, care.

Even if every person has food and housing, not everyone can decide the direction of a city, control a grid, or set the rules of a model. Power is relational and, in one sense, positional: a person's capacity to impose a decision exists relative to others. Technology can distribute tools, but it can also concentrate infrastructure in the hands of those who own data centers, chip factories, robots, and platform rights.

A post-scarcity economy can be democratic, oligarchic, or authoritarian. Free basic goods do not guarantee freedom. A state can offer comfort in exchange for conformity; a company can offer services in exchange for dependence. Legitimacy—the acceptance of rules as justifiable—remains scarce because it cannot be manufactured through efficiency alone.

When any voice and style can be imitated, provenance becomes infrastructure. Signatures, chains of custody, reputation, and verification institutions will not eliminate falsehood, but they will make accountability possible. In a world of infinite content, trust may become one of the most expensive forms of capital because it is built slowly and lost quickly.

Power over filters is as important as power over factories. Whoever decides what information is visible, which agent receives access, and which rules are applied can shape choices without explicit prohibitions. Algorithmic pluralism, interoperability, and a right to explanation are therefore components of resource distribution. To control the attention of others is, to some extent, to control their practical reality.

When automobiles become common, status shifts to brand, rarity, experience, or access. When images become free, the original and the artist's attention become valuable. When high-quality education can be delivered by AI, a selective institution may retain value through its network and social signal. Hirsch warned that the supply of positional goods is rigid: not everyone can be in the top 1%, at the center of the network, or at the exclusive table (Hirsch, 1976).

This means relative dissatisfaction can survive absolute abundance. People compare not only whether they have enough but also where they stand. Post-scarcity politics must reduce the dependence of dignity on rankings that everyone cannot win. It cannot eliminate ambition, but it can prevent access to health, political voice, and freedom from becoming positional prizes.

Some decisions consume opportunities that cannot be recovered. A natural area converted into a city, an industrialized landscape, demolished heritage, or a homogenized culture cannot be fully restored. An economy with enormous construction capacity can produce change faster than society can reflect. An abundance of action creates a scarcity of prudence.

This justifies slow procedures for irreversible decisions even when the rest of production accelerates. Not every delay is useless bureaucracy. Sometimes deliberation time is a safety technology. The challenge is to distinguish genuine verification from rent-seeking veto and inertia.

After this inventory, post-scarcity can be redefined more precisely. It is not the abolition of scarcity but the relocation of economic conflict away from material survival. People will continue to compete for attention, status, places, and influence. But they should not lose water, food, shelter, health, or access to knowledge when they lose that competition.

This version is both more modest and more radical. It is modest because it accepts limits and differences. It is radical because it requires much of the disciplinary power of labor markets and property to be withdrawn from the domain of elementary life.

Abundance may even intensify positional competition. As differences in material comfort narrow, people may invest more in subtle signals of exclusivity, authenticity, and proximity to centers of power. Virtual goods and algorithmic reputations do not eliminate status; they lower its production cost and accelerate its cycles. A politics of sufficiency must accept this dynamic without allowing social scores to become conditions for rights.

The scarcity of irreversible experiences includes the opportunity to be first, to live through a historic moment, or to meet a particular person. Copies and simulations can be valuable, but they are not identical to the event. This limit is healthy: a good world need not turn every difference into a production problem. Some things remain precious precisely because they are finite.

8. The Limits of Human Biology and Culture

Post-scarcity is not the end of human competition. It is the chance to detach competition from hunger, dependence, and the right to subordinate others.

8.1 A Plural Species: Egalitarianism, Accumulation, and Empire

The question of whether humans are “built for inequality” seems to demand a simple biological answer, but the formulation itself hides the problem. A species is not designed for a single institution. Natural selection did not produce a program called empire, democracy, or communism, but a set of predispositions, emotions, and strategies that work differently in different environments. Humans can cooperate with strangers, punish cheaters, share resources, and restrain leaders; they can just as readily form coalitions, pursue status, exploit asymmetries, and accept authority. What we call “human nature” is precisely this tense plurality.

Life in small groups favored mutual dependence. Hunting, child care, defense, and variable resources made sharing and a reputation as a good partner advantageous. But the same social life made position important: who is listened to, who has allies, who controls information, who can impose costs, and who is desired as a partner. Cooperation and competition are not two successive eras of humanity but two functions of the same social intelligence. Even generous acts can produce prestige, and rivalry itself can be channeled toward useful contributions.

Recent anthropology warns against idealizing societies called egalitarian. Stibbard-Hawkes and von Rueden show that we know of no societies in which equality is achieved across all domains. Differences appear in health and skills, social networks, leadership, gender, age, property, and reproduction. What distinguishes many egalitarian groups is not the absence of all differences but the fact that most members can retain access to resources, autonomy, and the ability to withdraw from abusive relationships. Equality is a process of limiting domination, not a photograph in which every outcome is identical (Stibbard-Hawkes and von Rueden, 2025).

This reinterpretation matters for post-scarcity. We do not need to prove that humans love perfectly equal distributions. We need to know whether they can sustain institutions that guarantee a material floor and prevent differences in talent, luck, or prestige from becoming arbitrary power. Historical evidence says they can, but not spontaneously and not forever. Sharing norms, rights of exit, transparency, competition among leaders, shaming of dominators, and redistribution were social technologies long before the modern state.

My intuition is that humans are not built for equality in the sense of identical outcomes, but they are built sufficiently for reciprocity, autonomy, and resistance to domination that a society of sufficiency is possible. At the same time, a minority strongly oriented toward control can have disproportionate effects if institutions give them leverage. A realistic political theory must be robust

both to ordinary self-interest and to the tail of the distribution: individuals who derive satisfaction from subordinating others, not merely from their own well-being.

Biology is therefore not a verdict but a distribution of risks and capacities. Culture selects which strategies become prestigious, institutions determine which strategies are profitable, and technology amplifies the power of those who control infrastructure. The same species can organize a band with recallable leaders, a hereditary empire, a cooperative, or an almost feudal digital platform. The fact that all are human does not make them equally inevitable or equally acceptable.

For most of the species' existence, humans lived in relatively small and mobile groups. This does not prove an original moral equality, but it constrained accumulation. Goods that were hard to transport had little value, perishable food was shared, and a dissatisfied individual could leave with allies. Leaders depended more on persuasion and reputation than on permanent coercive apparatuses. In some groups, braggarts were ridiculed, successful hunters could not claim all the prey, and major decisions required consensus. Christopher Boehm called such practices "reverse dominance hierarchies": the coalition of ordinary members keeps aspiring dominators under control (Boehm, 1999).

This picture must immediately be qualified. Mobility and sharing did not eliminate differences in influence, gender, age, health, knowledge, or reproductive success. Some hunter-gatherer populations had slavery, warfare, or powerful leaders; others were far more egalitarian. The newer framework proposed by Stibbard-Hawkes and von Rueden correctly insists that "egalitarian" does not mean equal outcomes across every dimension. Moreover, a recent study of the Hadza suggests that aversion to inequality is stronger when the difference is personally disadvantageous than when the person benefits (Smith et al., 2026). Self-interest does not disappear in sharing societies.

Agriculture changed the ecology of power. Land, animals, stores, and infrastructure could be controlled, defended, and inherited. Seasonal harvests created visible surpluses and points of fiscal capture. Sedentism made exit more costly, while population density enabled armies, administrations, and specialized classes. The comparative study coordinated by Borgerhoff Mulder found substantial intergenerational transmission and inequality in small-scale pastoral and agricultural societies, especially for material wealth. Archaeological evidence from housing indicates a general rise in disparities after the Neolithic, more pronounced in Eurasia, where large animals, the plow, and military mobility enabled accumulation and elite expansion (Borgerhoff Mulder et al., 2009; Kohler et al., 2017).

This tendency should not, however, be turned into an inevitable ladder from freedom to empire. Agricultural and urban societies varied enormously. Some states were more collective, with public investment, negotiated taxation, and more dispersed power; others were extractive dynasties. Comparative archaeological research on the distribution of power shows alternations and combinations, not a single direction (Feinman et al., 2026). Even within empires, cities, temples, guilds, communes, and rural communities could preserve different degrees of autonomy.

Several millennia of empires are nevertheless serious evidence. They show that large-scale domination can be stable, productive for elites, and compatible with complex social lives. Empires built roads, legal systems, and markets, but they did so alongside tribute, war, slavery, and control over reproduction. Their stability does not prove that people generally prefer subordination; it proves that coercive apparatuses, religious legitimacy, selective distribution of benefits, and large-scale coordination can sustain a system even when the majority would not have chosen it from a free menu.

Equally important, the modern era offers several centuries in which very rich states have reduced internal violence, expanded rights, taxed progressively, and universalized education, health care, or pensions. These institutions are fragile and imperfect, but they too are human products. Long history does not allow us to say that equality is the natural state. Nor does it allow us to say that empire is destiny. It tells us that the distribution of power responds strongly to the technology of accumulation, the cost of exit, coalition organization, and inheritance rules.

Post-scarcity changes all of these variables. Basic goods may become harder to use as instruments of coercion, while data centers, robots, and networks may become the new stores of controllable surplus. Digital mobility may facilitate exit, while surveillance may make exit nearly impossible. Abundance does not automatically return us to the egalitarianism of mobile groups; it can create a software-administered neo-empire. The result depends on who owns the productive stock and whether citizens can leave, refuse, and contest.

8.2 Dominance, Prestige, Bodies, and the Persistence of Hierarchy

The desire for status is not a single impulse. The literature distinguishes dominance from prestige. Dominance obtains compliance through the capacity to impose costs, intimidation, or control; prestige attracts deference because a person possesses skills, knowledge, or qualities from which others can benefit. Both can produce rank and influence, but the moral relationship differs: dominance takes; prestige is granted. Experimental and observational studies suggest that these are distinct pathways, although the same person can combine them (Henrich and Gil-White, 2001; Cheng, Tracy, and Henrich, 2010; Cheng et al., 2013).

This distinction answers a central fear about equality. If we remove the possibility of becoming superior by controlling vital resources, we do not eliminate the need for recognition. We can redirect it toward mastery, courage, generosity, discovery, art, athletic performance, care, or recallable leadership. A society without differences in prestige is probably impossible and perhaps not even desirable. A society in which prestige automatically converts into a right to coerce is dangerous.

Status has often had reproductive consequences. A meta-analysis of 288 estimates from 33 nonindustrial populations found a positive association between male status and reproductive success, with important variation by subsistence system and marriage regime (von Rueden and Jaeggi, 2016). More recent comparative work shows that human reproductive inequality is real but

not extraordinary relative to other mammals and is strongly reshaped by institutions, monogamy, contraception, and norms (Ross et al., 2023). Biology therefore provides one reason status has been pursued, but it does not determine a single social architecture.

In many empires and aristocracies, access to bodies was an explicit component of power: sexual slavery, arranged marriages, harems, unequal divorce rights, control over fertility, and elite impunity. Speaking of “access to bodies” requires precision. A person's body is not a scarce good to be redistributed. Consent places intimate relationships in a different category from food or housing. Material equality can reduce situations in which someone accepts a relationship because of hunger, debt, or dependence; it cannot create an obligation to desire someone.

This is precisely why some resistance to equality may be darker than the defense of property. For some, wealth is valuable not only for the comfort it buys but for the difference in power it creates: servants who cannot refuse, vulnerable employees, dependent partners, exclusive access, and public signs of superiority. If automation gives everyone a material floor, these levers weaken. A person may remain rich but can no longer purchase obedience as easily. It is plausible that a minority will hate post-scarcity precisely because it reduces the relational value of wealth.

This motive should not be generalized to everyone who accepts inequality. Many sincerely believe that differences reward contribution, risk, effort, or responsibility. Starmans, Sheskin, and Bloom argue that people often prefer distributions perceived as fair rather than arithmetical equality; when equality and merit appear to conflict, they may prefer an inequality they consider just (Starmans et al., 2017). The problem is that perceptions of merit are easily manipulated and tend to ignore inheritance, luck, monopoly, and bargaining power.

A stable post-scarcity order should not promise equality of desire or admiration. It should guarantee genuine consent, prevent the purchase of vulnerability, and break the link between success in one domain and the right to control all others. An admired scientist does not automatically receive political power; a wealthy entrepreneur does not receive control over employees' privacy; a celebrity does not receive immunity. Pluralizing status and separating spheres of power are more realistic than abolishing rank.

The observation that some subordinated people would gladly trade places with those who subordinate them is, in my view, correct for a meaningful part of the population. It describes opposition to one's personal position, not necessarily to the hierarchical principle itself. Someone may consider the system unjust only because they are low within it. If offered a credible path upward, they may become one of its most energetic defenders. Experimental literature frequently finds an asymmetry between aversion to disadvantageous and advantageous inequality; the recent Hadza result points in the same direction without implying that all preferences are selfish (Smith et al., 2026).

The first mechanism is aspiration. Hierarchy offers a story: the low position is temporary, and following the rules may be rewarded. Even when objective mobility is small, visible examples of people who have risen sustain hope. The status lottery is attractive because the prize is large and

because the alternative, collective contestation, is risky. Meritocratic culture can transform inherited advantage into evidence of personal worth and structural failure into individual blame.

The second mechanism is system justification. A stable order provides moral, religious, or technical explanations for differences and reduces the anxiety created by the idea that the world is arbitrary. The meta-analysis by Essien and colleagues, covering more than 700,000 observations, found in some domains implicit favoritism toward the advantaged group among disadvantaged groups, but explicit preferences were often neutral or favored the in-group, with substantial heterogeneity across contexts (Essien et al., 2021). Another meta-analysis suggests that system justification can have a psychologically palliative effect for disadvantaged minorities, sometimes reducing distress, but can become harmful when it entails internalizing inferiority (Lima et al., 2025).

The third mechanism is delegated domination. Pyramid systems offer almost every level a lower one: the supervisor dominates the worker, the citizen the migrant, the adult the child, the majority the minority. Small power received in exchange for large obedience can stabilize hierarchy. Empires used local intermediaries, auxiliary armies, officials, and co-opted elites precisely to distribute the advantages of domination without distributing the center of power.

The fourth mechanism is cultural learning. Children observe who is listened to, who is served, and which behaviors are rewarded. If prestige is associated with cruelty, ostentation, and impunity, those strategies become aspirational. Institutions do not merely reflect preferences; they train them. A labor market in which the abusive manager is promoted produces more abusive managers. A society in which generosity and competence bring prestige can produce a different cultural selection.

There is also a more benign explanation. Hierarchies can coordinate quickly, divide responsibilities, and reduce negotiation costs. A subordinate may accept a leader not because they love submission but because collective decisions are difficult and skills differ. The problem arises when a functional hierarchy becomes hereditary, opaque, and impossible to contest. Not every difference in authority is exploitation; not every obedience is system justification. The criteria are limited purpose, relevant competence, possibility of recall, right of exit, and absence of control over the private sphere.

Counterexamples are equally important. People show solidarity, accept costs to punish injustice, refuse undeserved advantages, and build anti-hierarchical movements. The history of abolition, trade unions, civil rights, and democratization cannot be explained by a desire merely to change places. The vicious circle exists, but it is not closed. It can be interrupted when people have material security, information, autonomous organizations, and prestigious examples of power exercised without domination.

My assessment is therefore intermediate. We cannot build post-scarcity on the assumption that beneficiaries will automatically become egalitarian. Some will try to recreate ladders of domination, while others will defend hierarchy in the hope of climbing it. But neither should we assume that the

majority has a biological need to exploit. Preferences are mixed, contextual, and institution-sensitive. Politics must make dominating strategies costly and prestige-based strategies rewarding.

Personalities differ, and part of these differences has a genetic component. Meta-analyses of twin studies estimate moderate heritability on average for psychological traits, including roughly 40% for personality overall, with variation across models and methods (Polderman et al., 2015; Vukasović and Bratko, 2015). A study of 1,987 Norwegian twins estimated heritable components for the two dimensions of social dominance orientation and found genetic overlap with political attitudes favoring hierarchy (Kleppestø et al., 2019). Later research indicates a partly shared genetic basis for authoritarianism and dominance orientation beyond Big Five traits (Kleppestø et al., 2024).

These findings are easy to misinterpret. Heritability does not mean that 40% of a person's personality is "genetic." It describes the proportion of observed variation among people, within a population and environment, that is statistically associated with genetic differences. If the environment becomes more uniform, heritability can rise even when interventions are highly effective; if the environment changes, estimates can change. A heritable trait can be shaped by education and institutions, while a trait with low heritability can be difficult to change.

More importantly, there is no single biological variable called "love of inequality." Social dominance orientation measures preferences regarding hierarchies between groups, not all forms of ambition, leadership, or competition. Genes influence dispositions through development, perception of the environment, temperament, and selection of situations; culture provides the categories and targets. The same aggressiveness can be channeled toward protecting a group, sport, entrepreneurship, or abuse. The same sensitivity to status can produce the prestige of a scholar or the domination of an arbitrary boss.

The maintenance of variation makes evolutionary sense. Strategies can be frequency- and context-dependent. In a violent environment, vigilance, coalitions, and dominance can bring advantages; in a cooperative group, reputation, generosity, and competence may be more profitable. A population composed only of dominators would suffer permanent conflict; one completely lacking defense would be vulnerable. Selection can preserve a distribution of temperaments without any one of them becoming a universal social destiny.

Post-scarcity will not erase this distribution. It is unlikely that all people will become peaceful, creative, and content once they receive basic services. Some will continue to pursue others' submission, convert prestige into monopoly, and test limits. That is precisely why institutions must be designed for adverse cases, not for the average person. Separation of powers, auditing, rights of exit, dispersed ownership, and the impossibility of buying impunity are the social equivalent of engineering redundancy.

There is also the risk of a new political biologization. If gene editing and embryo selection become more powerful, elites could present their advantages as biological superiority or try to select

traits of obedience and competitiveness. Predictions here are highly speculative, and social traits are polygenic and environment-dependent. But the principle should be established early: biological difference confers no right to dominate, and enhanced capabilities do not justify a hereditary caste.

The correct conclusion is not that biology does not matter, but that it matters in a way incompatible with simple determinism. We have reason to expect persistent variation in appetite for power and equality. We have just as much reason to believe that the institutional environment determines which individuals reach leadership, which strategies are imitated, and how much harm an extreme of the distribution can cause.

8.3 Motivation, Boredom, and Abundance for Real People

The fear that abundance will produce apathy begins from a real fact: much of the structure of modern life is supplied by necessity. Work sets the hours of the day, provides roles, colleagues, goals, and a measure of social worth. When this structure disappears abruptly through unemployment, the psychological effects are often negative. Meta-analyses find deterioration in mental health during unemployment and improvement after reemployment (Paul and Moser, 2009). But involuntary unemployment is not equivalent to material freedom: it combines loss of income, status, routine, and control within a society that still defines worth through employment.

Research on motivation offers a more optimistic picture. Self-determination theory distinguishes autonomous motivation from behavior controlled through reward and punishment. People tend to persist and feel well when activity supports autonomy, competence, and relatedness. The classic meta-analysis of 128 experiments showed that some extrinsic rewards can reduce intrinsic motivation, especially when perceived as controlling instruments (Deci, Koestner, and Ryan, 1999). Economic necessity can generate effort, but it is neither the only nor the best source of engagement.

Experiments with unconditional income do not show a mass exodus from work. In a U.S. experiment with 1,000 low-income people, the transfer reduced working time by roughly one to two hours per week, and partners also modestly reduced work; participants used some of the time for job search, education, or other activities, without uniformly positive effects (Vivalt et al., 2024). The Finnish experiment did not produce a clear increase in employment, but neither did it produce a collapse; participants reported greater well-being in surveys, although the lack of a baseline measurement limits causal inference (Kangas et al., 2020). These programs are far from post-scarcity, but they contradict the simple scenario in which modest security destroys the will to act.

The risk of boredom is nevertheless real. Boredom is not the absence of tasks but the inability to engage in an activity that feels meaningful. It is associated with depression, impulsivity, and aggression, and experiments by Pfattheicher and colleagues found a link between boredom and sadistic behaviors in certain contexts (Danckert and Merrifield, 2018; Pfattheicher et al., 2021). A

society that offers infinite entertainment but little autonomy and few real consequences can produce a toxic combination of passivity and shock-seeking.

“Dark” motivations do indeed supply energy: competition, envy, desire for revenge, fear of falling, and the need for superiority have built companies, empires, and careers. Their complete disappearance would change social psychology. But intensity should not be confused with value. A system can extract enormous productivity from fear and humiliation just as a machine can be driven beyond its limits. The question is not whether dark motivations work, but what costs they produce and whether more stable substitutes exist.

Sport, games, art, research, exploration, care, community building, and difficult projects show that people seek challenges even when survival does not depend on them. The mountain is climbed precisely because it is difficult, not because of a salary. Games artificially create scarcity, rules, and feedback; science creates cognitive frontiers; communities create obligations and gratitude. Post-scarcity does not eliminate the task of producing meaning. It moves that task from employer and market toward the person, community, and culture.

Not everyone will manage this freedom well. Some will prefer routines supplied by institutions, others will alternate periods of creation and withdrawal, while a minority may drift into addiction, simulations, or aggression. The solution is not to reinstall hunger as a motivational instrument, but to build options with real structure: lifelong education, voluntary or rewarded civic service, competitions, mentoring, local projects, care roles, and a right to participate in decisions. Freedom without capacity and belonging can become emptiness; freedom with competence and relationships can produce a life more intense than compulsory work.

My verdict is that aggregate motivation will not disappear, but it will be redistributed and become less predictable for organizations. Some unpleasant economic activity will need to be automated or rewarded much better. Meaningful activities will attract people without coercion, while meaningless ones will lose their workforce. This is not a problem of post-scarcity; it is one of its moral benefits and one of its industrial tests.

A robust post-scarcity society should not be designed for an altruistic “new human” or for the economic caricature of the exclusively selfish individual. It must work for a varied population: cooperators, competitors, introverts, leaders, vulnerable people, creators, opportunists, and a dominating minority. The design principle is the same as in computer security: you do not assume every actor is malicious, but you limit the damage a malicious actor can cause without paralyzing the system.

The first element is a universal floor, not equality of all outcomes. Food, water, decent housing, health, education, connectivity, basic energy, and access to mobility should be treated as infrastructures of freedom. People can earn more, build rare things, and pursue prestige, but they cannot use control over necessities to obtain obedience. This floor transforms consent: the employee can refuse, the partner can leave, the citizen can criticize without losing survival.

The second element is separation of spheres of power. Economic wealth should not buy unlimited political influence, judicial access, privacy, or informational control. Scientific prestige should not confer universal moral authority. A managerial role should not permit control over private life. A society can tolerate local and temporary hierarchies if they are limited to relevant competence, transparent, contestable, and recallable.

The third element is the pluralization of prestige. If there is a single social ladder dominated by money, every material difference becomes a difference in human worth. A more stable culture recognizes multiple forms of excellence: knowledge, art, care, sport, civic courage, craftsmanship, problem-solving, protection of nature, and the ability to develop other people. Prestige does not disappear, but competition is no longer a single game in which most people must lose.

The fourth element is ownership. If a small number of entities own the models, robots, networks, and energy, they can provide universal services and still dominate social life. Public ownership, cooperatives, citizen funds, open standards, interoperability, and competition are complementary instruments. Neither a single state nor a private monopoly offers sufficient protection. Power must be dispersed across levels and forms of ownership.

The fifth element is deliberate cultivation of meaning. Education cannot end when AI knows the answer; it must become training in judgment, taste, autonomy, and the capacity to choose projects. Communities need rituals, responsibilities, and spaces in which contribution is visible. Challenges can be voluntary without being trivial. An expedition, a research project, restoration of an ecosystem, or caring for a person is real even if no one starves for refusing.

The sixth element is protection of exit. The practical egalitarianism of many small groups depended on the ability to leave. In a digital society, exit requires portability of identity and data, interoperability, access to alternative infrastructure, and the right to function without continuous surveillance. A free service from which you cannot leave is dependence, not abundance.

These principles will not eliminate conflict. There will be people who regard every tax as theft, every universal floor as a reward for laziness, and every limit on wealth as an insult. There will also be movements that confuse equality of dignity with uniformity of tastes and outcomes. A stable order must allow experimentation and difference while preserving the anti-domination red line: no one may convert an advantage into a right over another person's body, voice, or survival.

The chapter's conclusion is deliberately cautious. Biology and culture do not make post-scarcity impossible, but they do make impossible the naive version in which technical abundance automatically cures power. There will remain struggles for status, meaning, and control. The civilizational gain would be to move those struggles above a safe floor and constrain them through rules that protect freedom. We do not need to eliminate competition; we need to eliminate the right to turn victory into unlimited subordination of the defeated.

9. How Fast Can “Enough” Be Built?

9.1 Why There Is No Single Date: The Road to 2050

The question “when do we reach the maximum?” assumes a single curve. In reality, there are several maxima. A geological maximum tells us how much material can be mobilized. An energy maximum tells us what flow can be produced without unacceptable thermal and ecological effects. An industrial maximum tells us how many mines, factories, and grids can be built in a decade. A social maximum tells us how much change can be absorbed without revolt, loss of skills, or capture. A welfare maximum may lie below the production maximum: beyond a point, more objects do not create a better life.

This book does not seek civilization’s thermodynamic maximum. It seeks the plausible speed to a sufficient threshold. For that purpose, each sector is evaluated through four variables: distance from the threshold, observed growth rate, stock-replacement time, and bottleneck risk. Solar energy can grow at double-digit rates annually, but buildings last for decades. Software updates weekly, grids expand over decades, and institutions can preserve old rules for a century.

For the timeline, we use a direct rule: a stage is reached only when all indispensable components of the service operate at the required scale. Rapid progress in one domain does not compensate for an absolute bottleneck in another.

In practice, the pace of transition is determined by the indispensable component that advances most slowly. Cheap energy without grids, robots without service capacity, or production without social access do not form a post-scarcity system.

For timing, what matters is whether every indispensable component—energy, materials, water, food, robotics, distribution, and institutional access—has actually crossed the threshold of universality, resilience, and sustainability. A region with energy and food but without safe water for a significant share of the population is not post-scarcity; neither is one in which services disappear with the job.

The components reinforce one another: energy supports water and materials, robots accelerate infrastructure, and governance can reduce risk or block investment. This caution prevents extrapolating a single technological curve—for example, assuming that a highly capable AI model instantly produces abundance.

I construct three scenarios. The slow scenario assumes geopolitical fragmentation, uneven investment, repeated climate crises, reactive regulation, and concentrated ownership. Productivity rises, but much of it is consumed by security, reconstruction, and rents. The baseline scenario assumes geopolitical competition without systemic war, continued cost declines in energy and automation, imperfect climate policy, and gradual expansion of social protection. The rapid scenario assumes strong industrial coordination, open standards, accelerated construction of grids and housing, designed-for recycling, use of AI for research, and a political compact that distributes part of productivity as universal services.

Table 7. Three Scenarios for the Transition to Basic Abundance

Dimension	Slow	Baseline	Fast
Global power in 2100	25–40 TW	50–80 TW	80–150 TW
Mature domestic robot	2060–2080	2045–2060	2038–2050
Post-scarcity threshold in wealthy states	After 2060, incomplete	2040–2060	2035–2050
Global threshold	After 2100 or fragmented	2070–2100	2055–2075
Institutions	Concentration and protectionism	Mixed universal services	Productivity dividends and open infrastructure
Ecology	Repeated overshoot of limits	Slow stabilization	Circularity and protection before consumption growth

Source: author’s integrated scenarios, not institutional forecasts.

The rapid scenario does not assume magic. It assumes that technologies already visible are deployed close to administrative and financial limits. The slow scenario does not assume technical stagnation; even there, AI and robotics advance, but access and public infrastructure lag behind. The difference between scenarios is primarily institutional.

During this interval, AI models become increasingly capable agents for code, research, design, administration, and education. The cost of standard cognitive tasks falls sharply. Data centers roughly double electricity consumption by 2030 but remain below 3% of global electricity in the IEA central scenario. Renewable capacity continues to grow rapidly, while grids, transformers, and permitting become the main constraints.

Industrial and logistics robotics expand; humanoids appear in factories and controlled warehouses, while early domestic products use teleoperation. Modular construction and autonomous machinery reduce costs in standardized projects. Critical minerals experience volatile prices, and copper and refining can delay electrification. Global food remains sufficient in aggregate, but conflict and climate create regional crises. States that channel productivity into public services can substantially reduce the cost of education, administration, and basic diagnosis.

There is no global post-scarcity in 2035. “Islands of abundance” may nevertheless appear: cities or states with very cheap electricity during certain hours, automated transport, universal AI access, and rapid construction. The risk is that these islands function by importing materials, embodied energy, and cheap labor from other regions, exporting scarcity outward.

In the baseline scenario, global electricity grows strongly, transport and part of heating are electrified, and storage and transmission capacity become much larger. Useful final energy may reach the equivalent of 30–45 TW, with regional differences. The first large cohorts of batteries and equipment reach recycling, but demand for new material remains high.

Robots become common in mines, agriculture, ports, warehouses, assistive care, and standardized construction. A USD 10,000–25,000 domestic robot can perform several useful hours per day, but it is not yet a worry-free appliance for every household. Subscription services and shared fleets grow faster than ownership. Some administrative and professional work is automated almost completely at the first-draft level, while verification becomes a central occupation.

Rich countries that build housing, energy, and universal services can reach a basic post-scarcity threshold around 2045–2055. This does not mean the disappearance of wages or prices, but that the essential bundle is guaranteed and very cheap. Other countries may have similar technology but limited access because of debt, conflict, or concentrated ownership.

9.2 2050–2100: Scenarios, Probabilities, and the Maximum Realistic Speed

By mid-century, the world population approaches 9.7 billion and its growth rate slows. The stock of clean infrastructure is large enough for recycling to matter systemically. Robots can work in variable environments with reliability comparable to complex appliances, while buildings and objects are designed for robotic maintenance. The cost of a standard hour of physical work can fall below human labor cost in most formal sectors.

In the baseline scenario, several states provide universal bundles of energy, water, connectivity, AI education, preventive health care, local transport, and basic housing. Some use dividends from automated capital, sovereign funds, or public ownership of infrastructure. Others tie access to employment, scores, or private platforms, producing conditional abundance.

Globally, gaps may remain. Countries with young populations and insufficient infrastructure need large construction volumes. Regions exposed to heat, drought, and sea-level rise consume resources on adaptation. Water conflict and migration can delay progress. The 2050–2070 window is nevertheless the first in which a universal material floor becomes industrially credible, not merely physically possible.

During this period, population reaches its projected peak and begins to stabilize or decline. If clean energy reaches 50–80 TW in the baseline scenario, each person benefits indirectly from an average 5–8 kW in a system much more efficient than the fossil one. Desalination, advanced recycling, precision agriculture, and robotic production can cover basic needs with large margins. Fusion, advanced geothermal, or space resources may contribute, but they are not indispensable.

At this point, the absence of global post-scarcity would become increasingly difficult to explain by physics. It would reflect war, capture, exclusion, ecological degradation, or a deliberate refusal to guarantee access. Humanity could have the capacity to produce enough while artificially preserving dependence on work and scarcity for social control. History shows that institutions do not automatically relinquish mechanisms of power when they become technically unnecessary.

The transition will be geographically fragmented. Some cities and states may provide almost free basic services while regions affected by conflict, climate, or weak institutions remain deprived. The gap between “the technology exists” and “the service is universal” can be several decades. Classical international aid will not be enough; regional factories, fiscal capacity, resilient infrastructure, and access to open technological standards will matter.

By the end of the century, the limit may become predominantly political. If energy and automation are mature, persistence of basic deprivation will resemble deliberate exclusion more than impossibility. This is why the global timeline has greater uncertainty than the technical one: it depends on wars, states, and ownership distribution, variables that do not follow regular learning curves.

The following ranges represent my subjective assessment, conditional on knowledge available in 2026. “Rich economies” means a significant group of high-income states, not every state. “Global” means effective access for at least 95–99% of the population to the basic bundle defined in this book.

Table 8. Subjective Probability of Reaching the Basic Post-Scarcity Threshold

Horizon	More high-income economies	Global scale
2050	35–55%	10–20%
2070	65–85%	35–60%
2100	85–97%	55–80%

Source: author's estimate, conditional on avoiding systemic war, major ecological collapse, and extreme concentration of productive infrastructure.

For 2050, I estimate a 35–55% probability that several rich economies reach a basic post-scarcity threshold and 10–20% for global attainment. For 2070, the ranges are 65–85% and 35–60%. For 2100, I estimate 55–80% for a global threshold. The probability does not reach 100% because war, authoritarianism, severe climate change, pandemics, technological accidents, and institutional degradation can dominate technical progress.

These estimates are more cautious than an extrapolation of AI and more optimistic than an extrapolation of current politics. The reason for optimism is the scale of energy potential, population stabilization, and the relatively small mass required for decent standards. The reason for caution is that universal access requires construction in the most difficult regions and a change in the distribution of power.

A sustainable terrestrial scenario of 50–100 TW appears sufficient for material abundance far above the basic threshold. At 100 TW and 10.3 billion people, average flow is 9.7 kW per person. With electrification and circularity, this could support comfortable housing, extensive mobility, abundant computation, regional desalination, advanced industry, and robotics. Not all energy would be consumed in homes; most would support infrastructure and production.

I do not recommend maximizing toward hundreds of terawatts merely because it may be possible. At 500 TW, average waste heat approaches 1 W/m²; at 1,000 TW, nearly 2 W/m². Land, materials, and local effects become severe earlier. A mature civilization should pursue a plateau of well-being, not infinite growth in material throughput. Additional expansion could partly move into space, where energy and heat can be managed differently.

The probabilities in this book are conditional on a world that avoids systemic disasters. They do not include a complete model of nuclear war, engineered pandemics, institutional collapse, or extreme climate change. They should be read as assessments of the capacity of a functioning civilization, not as assurances that civilization will remain functional.

The relevant “maximum” is not ultimate physical production but the level at which additional material consumption contributes little well-being and imposes large ecological costs. Once comfort, health, and freedom are secured, progress can shift toward time, knowledge, art, relationships, and exploration. A mature civilization is not the one that extracts the most, but the one that transforms few resources into many human possibilities.

Four developments could move the baseline scenario roughly ten years earlier. The first is robust robotics that dramatically lowers the cost of building power plants, grids, mines, housing, and factories. The second is cheap storage across multiple durations, removing bottlenecks in integrating variable energy. The third is automated design of materials and processes that reduces demand for copper, cement, and energy. The fourth is a distributive institution—dividends, universal services, public funds—adopted early and copied rapidly.

Cheap commercial fusion would help, but probably after the other transformations are already under way. Artificial intelligence far beyond the human level could accelerate research, but it cannot arbitrarily compress the time required for concrete to cure, permits to be issued, ecological systems to recover, or trust to form. Even a superintelligence must move matter through the world.

A war among major powers could destroy semiconductor and mineral supply chains, redirect robots toward weapons, and consume decades of capital. Climate change could turn development investment into reconstruction investment. Monopolies could slow diffusion to preserve margins. Cyberattacks could trigger a political backlash against autonomy. A badly managed labor transition could produce polarization and technological blockage.

More subtly, partial success could delay complete success. If elites obtain secure enclaves, their own energy, robots, and advanced health care, their pressure for universal services may decline. A society can become post-scarcity for 10% and more coercive for everyone else. This is not an impossible accident but one of the central risk scenarios.

The transition is driven by systems that must advance together. Cheap energy without grids, robots without maintenance, AI without data, and institutions without implementation capacity do not produce sufficiency. The pace is often set by the essential component that changes most slowly, but no equation is needed to express this. It is an engineering observation: a service works only when all of its mandatory links are available.

The strongest accelerator would be using AI and robotics to build the transition's own infrastructure: grid design, faster permitting without weaker oversight, mineral exploration, recycling, modular construction, and predictive maintenance. The greatest risk is the opposite loop, in which monopolies and conflicts slow investment while the population loses trust in technology because benefits are distributed unfairly.

10. The Personal Robot: The Automobile of the Twenty-First Century?

10.1 From the Automobile to the Robot: Product Generations and Full Cost

The question of whether we will have personal robots “as we now have cars” is useful because the automobile combines an expensive product, public infrastructure, financing, insurance, maintenance, and status. It did not become universal in a day, and even today it is not owned by every person. In many countries, households own one or more cars; elsewhere, access comes through public transport, taxis, motorcycles, or sharing. Diffusion has depended on income, urban form, industry, and policy.

The personal robot will probably resemble the automobile more than the phone. It has moving parts, physical risk, wear, and a need for service. It may cost thousands or tens of thousands of dollars. Its value depends on software infrastructure and parts. But it has one decisive difference: the automobile generally consumes the owner's time in order to function; the robot produces time for the owner. If it can work autonomously, it can justify a higher cost than an ordinary appliance.

For this analysis, a personal robot need not be humanoid or conscious. It is a mobile physical system that can perform, without continuous supervision, a portfolio of tasks in a household environment: cleaning, transporting objects, simple food preparation, loading appliances, safety monitoring, mobility assistance, inventory, and light maintenance. It must learn preferences, ask for help when uncertain, and stop in a safe state.

The minimum economic threshold could be two useful hours per day, roughly 730 hours per year. At an annual cost of USD 2,760, that is USD 3.78 per hour, without assigning value to nighttime availability or assistance. At four hours per day, the cost falls below USD 2 per hour. If the robot requires one hour of human intervention for every three autonomous hours, the economics change. Marketing metrics about “tasks” are therefore insufficient; we need net useful hours, interventions, incidents, and service cost.

The first generation, 2026–2035, is an early-access product. The price is USD 15,000–50,000 for complete systems, while cheaper models are platforms or have restricted capabilities. Teleoperation is common. Wealthy households, researchers, and people with special needs are the first customers. Value comes from experimentation and a few tasks, not from general replacement of domestic labor.

The second generation, roughly 2035–2047, becomes a complex appliance. Prices of USD 8,000–20,000 are plausible, financing and servicing are standardized, and the robot provides two to four useful hours per day in minimally adapted homes. Insurers, care providers, and residential buildings may subsidize purchase because it prevents falls, reduces hospitalizations, and supports independent living.

The third generation, roughly 2045–2060, becomes a mass-market product. A USD 3,000–10,000 platform, with modules and subscriptions, can be purchased like a premium washing machine or low-cost vehicle. General autonomy is sufficient for most repetitive tasks, and teleoperation is exceptional. Some households own several specialized units; others use a shared building robot.

These intervals overlap. Cheap, limited products may appear early; reliability in untidy homes may lag. Asian markets with scaled manufacturing and aging societies may adopt faster than countries with cheap domestic labor.

“Can afford” can mean direct purchase, financing at less than 10% of household income, subscription, access through a public service, or justification by income generated. Using the annual cost model, an early USD 20,000 robot costs about USD 6,460 per year. For this to represent 10% of income, the household needs about USD 65,000 annually. A mature USD 10,000 robot, at USD 2,760 annually, corresponds to income of about USD 28,000. A mass-market USD 5,000 product, at USD 1,270 annually, becomes compatible with income of USD 12,700 if financing and service are available.

These thresholds are not direct comparisons across countries because purchasing power, local costs, and informal incomes differ. They show that a mature robot could become affordable to a large share of the global middle class, but not automatically to the poorest households. Global inequality remains extreme: the World Inequality Report 2026 estimates that the bottom half receives less than 10% of global income and owns roughly 2% of wealth, while the top 10% owns three quarters of wealth (World Inequality Lab, 2026). Hardware prices can fall without democratizing finance or ownership.

A domestic robot should be evaluated more like a medical device and a vehicle than a gadget. It moves near children, older people, animals, fragile objects, stairs, fire, and water; it handles keys, cameras, medicines, and intimate information. An impressive success rate in a demonstration is not enough if the rare error causes a serious accident or every blockage requires operator intervention. Commercial cost will include certification, insurance, cybersecurity, updates, parts, local service, consumables, and legal liability. The gap between a ten-thousand-dollar prototype and a trusted household helper may be larger than the gap between the prototype and its physical mechanism.

Homes themselves will need adaptation. Thresholds, doors, furniture, outlets, packaging, and storage are designed for human hands and perception. Standardizing the environment can make the robot cheaper more effectively than endlessly perfecting the robot: easy-to-grip handles, markers, compatible objects, safe charging zones, and common interfaces. The first real savings may appear

in buildings, hotels, hospitals, and neighborhoods designed for fleets before every old house can host a general-purpose humanoid.

The automobile comparison remains useful for a social reason. Cars spread not only because engines became cheaper but because roads, gas stations, mechanics, credit, insurance, rules, and a parts industry emerged. The personal robot will need a similar ecosystem, though predominantly digital and electric. Access through service will therefore probably precede universal ownership: the provider can amortize the robot across multiple households, centralize maintenance, and combine autonomy with remote human assistance. For the user, available function matters more than legal title.

10.2 Ownership, Access, Functions, and the Pace of Adoption

A robot can be owned by the household, leased, provided by an employer, included in building rent, operated by a municipality, or summoned on demand. Each model changes utilization. Ownership provides availability, personalization, and privacy, but leaves the robot idle part of the time. A shared fleet increases utilization and lowers cost but introduces logistics, hygiene, waiting, and access to data from multiple homes.

For rare tasks—moving furniture, deep cleaning, repairs—on-demand service is efficient. For mobility assistance, security, and daily routines, a dedicated robot is more valuable. A mixed model is likely: a modest personal appliance and specialized robots called from a fleet.

Public services can accelerate access. A health system can provide robots to older people if the cost is lower than institutional care. A municipality can maintain fleets for cleaning, delivery, and interventions. Schools can use shared robotics laboratories. In this sense, “how many people have a robot” matters less than “how many have access to hours of robotic labor when they need them.”

The following table separates dedicated household ownership or leasing from broader access through services. The percentages are the author’s estimates, not market forecasts.

Table 9. Estimated Adoption of Personal Robots or Equivalent Access

Year	Households globally	Households in wealthy economies
2030	<1%	1–3%
2035	1–4%	5–15%
2045	8–20%	25–55%
2055	25–50%	60–85%
2070	50–80% access	75–95% access

Source: author’s estimate. “Access” includes ownership, subscription, shared fleet, building-owned robot, or on-demand service.

In the baseline scenario, fewer than 1% of the world's households have a dedicated general-purpose robot in 2030. In rich countries, penetration may be 1–3%, concentrated in early access and care. By 2035, I estimate 1–4% globally and 5–15% in rich countries. In 2045, 8–20% globally and 25–55% in rich economies. In 2055, 25–50% globally and 60–85% in rich countries. By 2070, useful access—not necessarily ownership—could reach 50–80% of the world's households and 75–95% in rich ones.

The rapid scenario moves these levels roughly five to ten years earlier. The slow scenario could produce millions of company-owned robots while global household ownership remains below 20% in 2055. This is an important possibility: robotization of production does not automatically imply democratization of the robot.

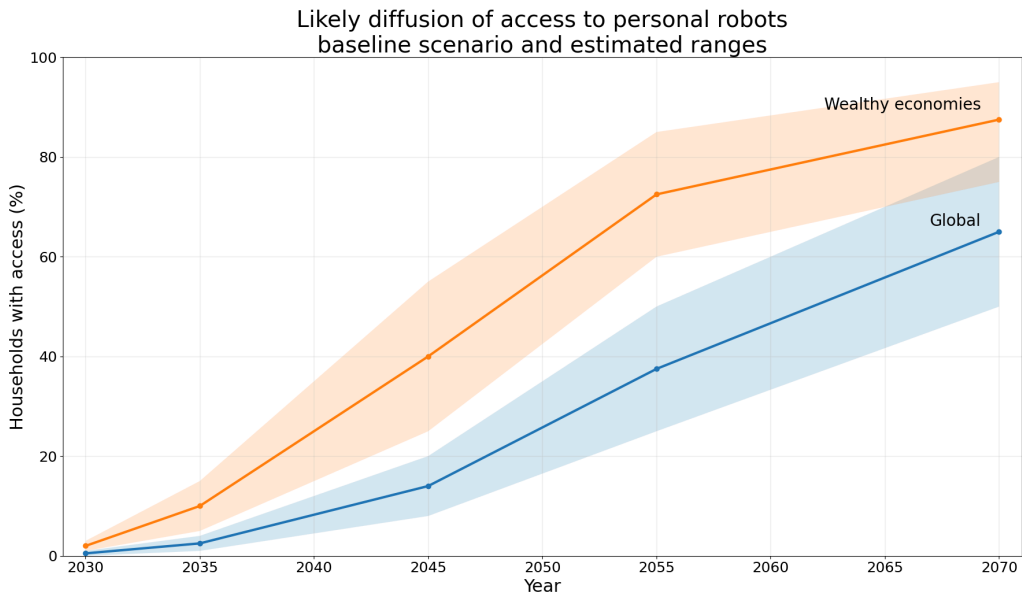


Figure 2. Baseline Trajectory and Estimated Ranges for Access to Personal Robots

Source: author’s estimate; ranges are not statistical confidence intervals.

The first common tasks will involve standardized objects and low risk: fetching things, carrying baskets, extended vacuuming, wiping surfaces, loading dishwashers, monitoring appliances, inventory, and patrol. Folding clothes and complete cooking are harder than they appear because they involve deformable materials, hygiene, knives, heat, and preferences. They can become feasible through environmental redesign: standardized clothes and baskets, modular kitchens, easy-to-grip containers, and appliances that cooperate with the robot.

This coevolution of the environment is probably underestimated. Automobiles changed roads, cities, and commerce. Robots will change handles, furniture, packaging, and housing standards. A “robot-ready” home can be easier to automate than reproducing human flexibility in every old house. The cost of intelligence can be reduced by intelligence in the environment.

Population aging and staff shortages create demand that is not merely about convenience. Robots can assist lifting, walking, hygiene, medication, meals, and emergency detection. Economic value includes preventing accidents and delaying institutionalization. In these cases, a USD 20,000 robot can be cheap relative to annual costs of intensive human care.

A care system can nevertheless use the robot in two ways. The augmentative mode retains staff and shifts physical effort and routine to the machine, increasing human relationship time. The substitutive mode cuts staff to a minimum, turning isolation into “efficiency.” Quality regulation must measure human outcomes, not only completed tasks.

Care is simultaneously a huge market and a moral test. A robot can lift a person, bring objects, monitor risks, and take over repetitive tasks, freeing time for human interaction. It can also become a justification for reducing contact, continuous surveillance, and lower-quality services for the poor.

The correct standard is not whether the robot replaces a human, but whether the person receiving care gains more autonomy, safety, and human relationship than before.

The first mass uses will probably combine limited tasks rather than deliver a universal butler. Cleaning, moving items within the home, simple manipulation, appliance monitoring, and physical assistance can create value before perfect conversation or general dexterity. Commercial progress will come from accumulated useful hours, not from anthropomorphic resemblance.

The personal automobile is largely a depreciating consumer good. A robot can be a productive asset: it can work for the household, be rented out, produce goods, or allow its owner to work elsewhere. This can accelerate financing, but also inequality. Households with capital buy robots that generate income; those without capital pay subscriptions or compete with them.

If robots become the property of a minority, the transition can shift income from labor to capital at unprecedented speed. A minimum wage does not protect a person whose labor is no longer purchased. The relevant instruments become distributed ownership, social funds, taxation of rents, dividends, and universal services.

Yes, personal robots are likely to become as common as automobiles in rich economies, but not before autonomy, servicing, and liability mature. The central window is 2045–2060 for most rich households and 2055–2075 for broad global access. A “robot in every home” is not necessary for post-scarcity; fleets and robotic infrastructure can provide useful labor more efficiently.

Nominal price has already entered the range of a cheap automobile. Functionality has not. The next ten years will test whether models can turn relatively affordable bodies into reliable hours of work. When that metric changes, adoption can become very rapid because the robot does not merely offer a new object but time—the scarcest personal resource.

For a household, the robot becomes affordable when it can avoid more expensive services, allow someone to remain at home, reduce domestic labor time, or generate income. For a company, the criterion is utilization: a robot working two hours per day has very different economics from one used across shifts. This initially favors fleets and shared services, where capacity can be moved among customers.

Mass individual ownership will come after standardization, local service, and predictable insurance. In rich economies, the personal robot may become comparable to a modest automobile in the 2040s–2050s; globally, access through services will precede ownership by a long margin. Benefiting from robotic labor does not necessarily mean having a humanoid in the living room.

11. The Politics of Abundance: Ownership, Access, and Power

11.1 Monopoly, Universal Services, and Ownership of Infrastructure

Technology can reduce the marginal cost of a good almost to zero while its price remains high. Property rights, monopoly, land, patents, regulation, and discrimination determine who captures the difference. Software is the obvious example: copying costs almost nothing, yet the license can be expensive. Robotically constructed housing can be cheap, but land and credit can make it unaffordable. Solar energy can be cheap at the plant gate, while the household still pays for the grid, debt, and rents.

Post-scarcity does not appear when technical cost falls, but when that decline is converted into a right of access. This conversion can be achieved through competitive markets, public services, cooperatives, common ownership, regulation, or combinations of these. There is no single correct institution for every good. Water, basic energy, and health have strong reasons for universal guarantees. Luxury and variety can remain markets.

High fixed costs and low marginal costs favor concentration. An AI model, robotic network, or automated factory requires major investment and can then serve millions of users. Network effects, data, and standards can make the leader increasingly powerful. If cognitive and physical infrastructure is controlled by a few companies, society can receive cheap goods and lose autonomy.

Monopoly is not only about price. It is the capacity to decide what is produced, what behavior is allowed, what data are collected, and who may compete. A robot platform can disable features, impose subscriptions, or refuse repairs. A post-scarcity society dependent on a provider's permission is fragile.

Open standards, interoperability, the right to repair, data portability, and the ability to operate locally are political infrastructure. They may look less spectacular than the robot itself, but they determine whether productivity becomes a common good or a rent.

The paradox of abundance is that a good with very low marginal cost can generate very high rents if the infrastructure producing it is concentrated. Software has already demonstrated this. Models, chips, robotic operating systems, and data markets can reproduce the same structure in the physical world. The cost of a service falls, but price remains high because the provider controls access and substitutes.

Competition is necessary, but not sufficient. Some infrastructures have economies of scale and network effects that recreate concentration. Open standards, portability, vertical separation, public

capacity, and cooperatives can preserve contestability. A post-scarcity economy cannot depend on the goodwill of a single provider, whether company or state.

Universal basic income is one possible solution for an economy in which work becomes scarce. But if housing, health, and energy have rigid or monopolized supply, monetary transfers can be absorbed by prices. An alternative or complement is a universal bundle of services: a quantity of energy, water, connectivity, mobility, education, health care, and housing guaranteed directly or through regulated options.

Universal services have the advantage of addressing the actual shortage and building capacity. Income provides choice and respects diversity of preferences. A robust architecture combines them: a service floor that cannot be lost and a monetary dividend from the productivity of automated capital. Above that floor, markets and preferences can operate.

If robots replace a large share of labor, the distribution of ownership over them becomes the equivalent of land distribution in agrarian societies or industrial capital in the nineteenth century. Three extreme models are possible. In the first, companies own almost everything and people buy services with wages or transfers. In the second, the state owns the infrastructure and distributes services, with risks of bureaucratization and control. In the third, households, cooperatives, and citizen funds own a significant share of the capital.

The plural model is probably safer. Natural-monopoly infrastructures—grids, water, certain foundational platforms—can be public or tightly regulated. Companies compete in products and services. Households can own robots. Sovereign or social funds can hold stakes in automated capital and pay dividends. Cooperatives can operate local fleets. Diversity of ownership reduces the risk that a single institutional failure will dominate.

Basic income can strengthen bargaining power and simplify social protection, but by itself it does not solve housing, health-care, or energy monopolies. If supply is rigid, the transfer becomes rent, insurance premium, or a higher price. Universal services and expansion of capacity must therefore accompany income. Sometimes it is more efficient to guarantee minimum energy, transport, education, and care directly than to distribute money into a blocked market.

Universal infrastructure need not mean a single provider. The state can finance the entitlement, while public services, cooperatives, and regulated firms deliver it. The criterion is that losing a job or a credit score must not interrupt access. Universality is a property of the outcome and the right, not necessarily of the organizational form.

Access policy does not require a single organizational model. Minimum energy can be guaranteed through a regulated utility, health care through social insurance and multiple providers, housing through public construction, cooperatives, supply rules, and targeted assistance, and cognitive tools through public infrastructure and interoperable private services. The common criterion is real ability to use the service, not the owner's label. A nominally free service that is scarce, humiliating, or impossible to contest is not universal; a competitive market can provide universality if the entitlement is funded and capacity is sufficient.

The order of interventions matters. Transferring money into a sector with rigid supply can raise rents; capping prices without investment can reduce supply; building at scale without transport and services can create dependent peripheries. Post-scarcity policy must begin with capacity: grids, housing, clinics, training, storage, repair centers, and redundant production. Only then does subsidy turn capacity into access. This is less spectacular policy than distributing a check, but it is the difference between nominal purchasing power and material possibility.

Plural ownership is a security measure, not merely an ideological preference. Exclusively private infrastructure can turn service into rent and the user into a dependent; exclusively centralized infrastructure can turn access into a political instrument. Public networks, competing firms, cooperatives, citizen funds, open standards, and the right to operate locally can constrain one another. In an automated world, freedom is guaranteed not only by income but by the existence of multiple paths through which a person can obtain a service, move their data, and refuse the authority of a single provider.

11.2 The Technology Dividend, Artificial Scarcity, and the Social Contract

Solar energy has no natural owner, but land, grids, and capital do. Collective data, public research, and educational infrastructure contribute to AI models, while profit can remain private. A “technology dividend” can be justified not as a penalty on innovation but as recognition of the social foundations of productivity.

Sources could include taxes on land rents, resources, carbon, monopoly rents, and large inheritances; public stakes in infrastructure; royalties for the use of public data; and returns from capital funds. Design matters. Taxing each robot as if it were a worker could slow the very productivity that is needed. It is more efficient to tax economic profit and rent than the act of automation itself.

Even in an abundant economy, people will work for additional income, status, meaning, mastery, and relationships. The difference is bargaining power. When survival does not depend on accepting any job, dangerous, humiliating, or meaningless activities must either be automated or paid and organized much better.

This could produce a more authentic labor market, not the disappearance of work. Scarce human activities—care, leadership, art, exploration, mediation, research, community building—may expand. Some will be paid; others will be voluntary contributions. A person may alternate between periods of project work, education, and caregiving without fearing the loss of access to basic life.

There is, however, a transition problem. Modern institutions tie income, identity, pensions, health care, and status to employment. If automation advances before these links are rebuilt, people may lose not only money but their social place. Policy must create roles and participation, not merely transfers.

The technology dividend can take the form of direct ownership, sovereign or citizen funds, taxation of rents, employee participation, and public services financed from productivity. Without such a mechanism, automation can increase output while simultaneously reducing the income of those who do not own capital. This is the scenario in which abundance of objects coexists with precarity of people.

Work will not disappear uniformly. Some activities will be automated, others will expand because they become cheaper, and others will be retained for reasons of trust and meaning. The transition requires portable benefits, continuous education, and the right to refuse degrading conditions under threat of poverty. The moral gain from automation appears only if the time released becomes people's time, not merely additional capacity for organizations.

Some industries maintain prices through restrictions on copying, incompatibility, obsolescence, and repair lock-in. In a world where technical costs fall, the temptation to create legal scarcity will increase. Subscriptions can turn purchased objects into revocable services. Digital rights controls can prevent local use. Very broad patents can control essential methods.

Not every form of exclusivity is illegitimate. Innovation requires financing, and creators must be rewarded. The question is duration and scope. For essential goods, society can buy patents, finance open research, impose compulsory licenses, or create public standards. Post-scarcity requires institutions that allow investment to be recovered without turning low marginal cost into a perpetual toll.

Renewable energy distributes the primary resource more evenly, but production chains can remain concentrated. Advanced chips, mineral refining, robots, and AI models become strategic assets. States may restrict exports, control software, and build technological blocs. Domestic abundance can be achieved through external exclusion.

A resilient global system needs regional redundancy, interoperable standards, and agreements around essential goods. Not every country must manufacture the most advanced chips, but the world should not depend on a single point of failure. Minimum capacity for water, food, energy, and emergency production should be distributed.

War is the largest consumer of post-scarcity. It destroys infrastructure, redirects research, justifies surveillance, and turns robots into weapons. No technological forecast can ignore peace as a productive resource.

Intellectual-property rights, incompatible licenses, repair lock-in, and closed standards can preserve scarcity after the technical problem has been solved. Some exclusivities stimulate investment; extended too long or too broadly, they turn knowledge into a permanent toll. The rule should be calibrated to the real cost of innovation and the public interest, not treated as an unlimited natural right.

Geopolitically, control of chips, refining, and robotic platforms could replace oil as a source of dependence. Complete autonomy for every country is inefficient, but dependence on a single bloc

is fragile. Diversification, alliances, common standards, and minimum regional capabilities are forms of resilience. Post-scarcity inside a single wealthy fortress is not stable in a fragmented world.

If citizens depend on systems they do not understand and cannot challenge, efficiency can erode democracy. Algorithms may allocate housing, energy, health care, and access to robots. Errors become political. Essential decisions require a right to operational explanation, appeal, audit, and a human alternative.

Democracy must also accelerate without surrendering control. Energy and housing infrastructure cannot wait decades in fragmented procedures, but communities cannot be treated as obstacles. Digital processes, simulations, transparent compensation, and clear deadlines can compress permitting. Legitimate speed is different from administrative force.

A minimal contract could have six components, presented here as a system rather than a list of slogans. First, a universal bundle of essential services, defined by outcomes and updated as technology changes. Second, plural ownership and a social fund that captures part of the return on automated capital. Third, open infrastructure: interoperability, repairability, portability, and local processing. Fourth, firm ecological limits, with prices and rules for carbon, water, land, and materials. Fifth, rights over time, privacy, and appeal against automated decisions. Finally, universal access to education and cognitive tools so that people can participate in governing the system rather than merely consume its outputs.

Table 10. Elements of a Social Contract for Abundance

Principle	Institutional mechanism	Risk it limits
Universal basic services	Guaranteed water, minimum energy, health care, education, connectivity, housing, and mobility.	Technical abundance alongside poverty.
Plural ownership	Competition, cooperatives, public infrastructure, and individual rights over data and agents.	Monopoly over robots and intelligence.
Open infrastructure	Interoperable standards, portability, auditability, and nondiscriminatory access.	Technological lock-in and permanent rents.
Ecological ceilings	Material, carbon, water, and biodiversity budgets, with prices and prohibitions where necessary.	Turning abundance into biophysical collapse.
Rights over time	Reduced working time, a right to disconnect, and distribution of productivity gains.	Replacing material scarcity with exploitation of attention.
Rights of contestation	Explanation, human appeal, traceability, and accountability for automated decisions.	Algorithmic power without legitimacy.

Source: the author's normative proposal.

No element is sufficient on its own. Income without housing can feed rent. Services without freedom can create paternalism. A market without limits can destroy ecosystems. Public ownership without transparency can concentrate power. The contract must preserve both the capacity to innovate and the freedom to refuse.

From the perspective of physics, the post-scarcity threshold is plausible. From the perspective of institutions, it remains open. Technology may reduce the cost of essentials faster than states reduce conditional access to them. In the most likely scenario, the world will pass through a contradictory period: rising technical abundance, economic insecurity for part of the population, and enormous wealth concentrated in automated capital.

This period is not inevitably the destination. It is the transition that must be governed. If productivity is converted into free time, services, and distributed ownership, post-scarcity can expand freedom. If it is converted into surveillance, dependence, and rents, it could produce the most comfortable form of political powerlessness in history.

Automation raises the constitutional stakes of ownership. When infrastructure can produce almost everything that is necessary, control over it begins to resemble governing power. Rights of access, algorithmic audit, transparency of decisions, and the ability to challenge them must be treated as political institutions, not product options. A subscription cannot replace citizenship.

The social contract of abundance should guarantee a material floor, disperse productive control, protect privacy, and preserve real spaces of choice. It does not promise equality of all wealth or prestige. It promises that nobody can use control over necessities, information, or robots to turn

another human being into a dependent with no exit. This is the political definition of “enough for everyone.”

Conclusion. Enough Is Not Infinite, but It Can Be Universal

The question with which we began can now receive a more precise answer. Yes, humanity appears to possess sufficient physical resources to provide a good material life for a population of roughly ten billion people. The solar flux is orders of magnitude larger than civilization's consumption. Mineral reserves and resources, together with substitution and recycling, are sufficient for basic infrastructure. Food production can grow moderately, while losses, diets, and distribution offer enormous room for improvement. Urban water can be treated and desalinated at a manageable energy cost. Artificial intelligence can lower the cost of expertise, and robots can gradually lower the cost of physical execution.

But this positive answer has four conditions. The first is to distinguish the service from the object. We cannot extend to ten billion people the material model of the richest, with very large homes, permanent individual mobility, rapid product replacement, and unlimited positional consumption. We can extend comfort, health, mobility, nutrition, education, and freedom if we produce them efficiently.

The second condition is speed. Energy systems, mines, factories, housing, grids, and water infrastructure are not software. Even with automated design, they have physical and political timelines. Humanity can reach the threshold, but not in five years and probably not in twenty at global scale. The first wealthy economies may reach it between 2040 and 2060. The realistic global window is 2070–2100.

The third condition is ecology. Material abundance cannot be built by increasing extraction by 60% by 2060, continuing record emissions, and replacing the biosphere with infrastructure. Clean energy and recycling are not sufficient if material demand grows without limit. We must stabilize material throughput and increase the service obtained from every tonne.

The fourth condition is distribution. Today, hundreds of millions of people go hungry and billions cannot afford a healthy diet, even as the system wastes more than a billion tonnes of food. Two billion lack safely managed water even though treatment technology is mature. These realities demonstrate that physical potential does not turn itself into a right. An automated world could multiply the same contradiction.

Absolute post-scarcity remains impossible. We cannot produce additional time for a person without extending life; we cannot give everyone the same position; we cannot replicate a unique place, a relationship, or a lost species. Even if thought, intelligence, and labor become cheap, attention, trust, privacy, security, ecosystems, desirable land, meaning, and decision-making power will remain scarce. Economics does not disappear. It shifts from managing shortages of objects to managing human and shared limits.

Personal robots are a useful symbol of this transition. In 2026, a humanoid body can cost as much as an inexpensive car, but it does not yet offer automobile-level reliability. Over the next two decades, useful autonomous hours will matter more than demonstrations. In the baseline scenario, robots become common in wealthy households between 2045 and 2060 and globally accessible through ownership or services between 2055 and 2075. Not every person will need one, just as not every city resident needs a car. But almost everyone will use robotic labor directly or through infrastructure.

The most important result of the analysis is not a year, a price, or a number of terawatts. It is a change in the nature of the question. Civilization does not appear condemned to manage inevitable material scarcity. It is condemned only to manage limits. We can choose intelligent limits: sufficiency, resilience, nature, time, pluralism. Or we can preserve artificial limits: hunger amid waste, empty homes beside homeless people, robots owned by a few, and intelligence available only conditionally.

In 2100, if the material threshold is not universal, the most likely explanation will not be that the planet lacked enough sunlight, metal, or knowledge. It will be that institutions found scarcity more useful than the freedom abundance could have made possible.

Appendix A. The Resource Landscape in 2026

Table 11. Synthetic Verdict on Resources for an Abundant Civilization

Domain	Physical potential	Dominant constraint
Energy	Very high; sufficient to multiply the current system.	Grids, storage, capital, permitting, and well-chosen land.
Metals	Sufficient overall; some supply flows are tight.	Mining lead times, refining, concentration, and local impacts.
Food	Sufficient in aggregate for human diets.	Access, conflict, losses, diet, soil, climate, and biodiversity.
Water	Urban needs are manageable energetically; agriculture depends on geography.	Local distribution, infrastructure, and water use in food production.
Land and biosphere	Not substitutable without limit.	Ecosystem protection and efficiency of material services.
Artificial intelligence	Rapidly becoming abundant for digital tasks.	Reliability, data, governance, and local energy cost.
Robotics	High potential, but slower maturation.	Open-world autonomy, service, safety, and standardization.
Institutions	Not a physical limit.	Distribution of ownership, legitimacy, and coordination capacity.

Source: integrated conclusions of the book.

The table must be interpreted across time horizons. Solar and wind energy have very large physical potential and are already scaling rapidly, but the system is constrained by grids and flexibility. Copper and some electrochemical materials have sufficient long-run potential, yet supply flows through 2035 are tight. Food is sufficient in aggregate, but access and ecological performance are poor. Water can be treated for urban needs, while agriculture remains dependent on geography and ecosystems. Robotics is abundant for narrow tasks, but not yet for general physical labor.

Appendix B. How the Estimates Were Derived

B.1 Available Solar Power

The solar power incident on Earth's disk is:

Incident solar power is obtained by multiplying measured irradiance at Earth's distance from the Sun by the area of the disk the planet presents to incoming rays. Using standard values gives approximately 174,000 TW.

With $1,361.6 \text{ W/m}^2$ and a mean radius of 6,371 km, the result is approximately 173,600 TW. Values used in the text are rounded to 174,000 TW. Capturing 0.01% corresponds to 17.4 TW; 0.03% to 52.1 TW; 0.1% to 173.6 TW. The calculation does not include atmosphere, night, efficiency, land use, or losses.

B.2 Comparing Energy Scenarios

For a population of 10.3 billion:

To compare scenarios, the total power of the energy system is divided by the projected peak population, approximately 10.3 billion people. The result is an average for the entire economy, not a household allocation.

20 TW corresponds to 1.94 kW per person; 30 TW to 2.91 kW; 50 TW to 4.85 kW; 80 TW to 7.77 kW; 100 TW to 9.71 kW; and 150 TW to 14.56 kW. The values include all uses across the economy.

B.3 Surface Area Required for Solar Collection

If one square meter of photovoltaic collector supplies an average of approximately 40 W after accounting for night, weather, and losses, an average power of 50 TW would require about 1.25 million km^2 of collectors. This represents approximately 0.84% of the planet's land area. Project footprints would be larger, and the real system would combine solar with wind, nuclear, hydroelectric, and geothermal power.

B.4 Why Waste Heat Matters Only at Very Large Scales

Spread across Earth's entire surface, the use of 100 TW would add an average of approximately 0.20 W/m^2 as heat, 500 TW nearly 1 W/m^2 , and 1,000 TW nearly 2 W/m^2 . These are not climate forecasts but order-of-magnitude calculations showing why heat dissipation becomes a limit at power levels far above what basic post-scarcity would require.

B.5 Order of Magnitude of Urban Desalination

At 100 liters per person per day and 10.3 billion people:

At 100 liters of urban water per person per day and a population of 10.3 billion, annual volume is approximately 376 km³. Desalinating all of it would be an extreme scenario, used only to show the order of magnitude.

At 3 kWh/m³, the energy requirement is 1,127.85 TWh per year. Relative to 28,600 TWh of global electricity in 2025, this is 3.94%. Intake, final treatment, pumping, storage, and losses are not included.

B.6 Why Water for Food Cannot Be Treated Like Urban Water

At 2,000–5,000 liters per person per day, the annual volume for 10.3 billion people is 7,519–18,799 km³. Hypothetical desalination at 3 kWh/m³ would require 22,557–56,393 TWh per year, approximately 79–197% of global electricity generation in 2025. The exercise is counterfactual: most water used to produce food comes from precipitation and soil moisture.

B.7 How the Full Cost of a Robot Is Estimated

To convert purchase price into annual cost, I used standard amortization accounting for service life and the cost of capital. The financial detail changes the values but not the conclusion: maintenance and actual utilization are decisive.

Annual cost combines the annualized share of purchase price, maintenance, software, energy, and insurance. I retained the same assumptions as in the table, stated directly to avoid implying more mathematical precision than the data support.

The early scenario assumes a purchase price of USD 20,000, seven years of use, financing at 8%, annual maintenance equal to 8% of purchase price, and separate costs for software, energy, and insurance. The resulting full cost is approximately USD 6,461 per year.

The mature scenario assumes a purchase price of USD 10,000, eight years of use, financing at 5%, maintenance of 6%, and lower operating costs. The resulting full cost is approximately USD 2,757 per year.

The mass-market scenario assumes a purchase price of USD 5,000, ten years of use, financing at 4%, maintenance of 5%, and standardized operating costs. The resulting full cost is approximately USD 1,266 per year.

These scenarios do not include taxes, catastrophic repairs, residual value, or the cost of teleoperation. They are comparative tools.

Appendix C. Scenario Assumptions

Table 12. Assumptions Separating the Slow, Baseline, and Fast Scenarios

Variable	Slow scenario	Fast scenario
Technology	Incremental progress, fragile robotic autonomy, expensive storage.	Rapid learning, standardized hardware, cheap storage and grids.
Geopolitics	Hostile blocs, trade wars, and infrastructure destruction.	Minimum cooperation on materials, climate, standards, and safety.
Climate and ecosystems	Reactive adaptation, large losses, and destabilizing feedbacks.	Rapid reductions, restoration, and protection before thresholds are crossed.
Ownership	Automated capital is highly concentrated.	Plural ownership, universal services, and productivity dividends.
Circularity	Short-lived products, hard to repair, weak recycling.	Durability, modularity, material passports, and advanced recovery.

Source: the author’s working assumptions for falsifying and updating the scenarios.

The scenarios are intentionally broad. They do not model every country and do not assign a single global rate. The fast scenario requires both technical progress and institutions capable of converting it into access. Very rapid technological growth under concentrated ownership may resemble the slow scenario for most of the population.

Appendix D. How the Predictions Should Be Read

The numerical estimates in this book have four degrees of certainty. Historical data and current observations are the most robust, although they may be statistically revised. Institutional projections, such as population or energy demand through 2030, use published models and scenarios. Order-of-magnitude calculations use transparent formulas and simplified assumptions. The author’s estimates of dates, probabilities, and robot adoption are speculative syntheses.

No estimate for 2070 should be read with the precision of an annual budget. Its function is to rule out two unjustified extremes: the promise that post-scarcity is only a few years away and the fatalism that physical resources make it impossible. The ranges identify a window in which infrastructure could be built if technological trends continue and institutions do not block diffusion.

A good updating criterion is the following. The timeline should move earlier if useful autonomous robot hours grow faster than total cost, if grids and storage expand without large cost increases, if critical materials are substituted or recycled at scale, and if more states convert AI productivity into universal services. It should move later if safety incidents block robotics, if clean energy remains trapped in interconnection queues, if shortages of copper and transformers persist, if climate damage destroys capital faster than it is built, or if ownership becomes more concentrated.

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About This Edition

This book is a prospective analysis, not a manifesto for unlimited consumption and not a deterministic prediction. It seeks to turn a debate dominated by utopianism and catastrophism into a falsifiable map: what resources exist, where the bottlenecks are, what remains scarce, and what events should cause us to update the timeline. This edition uses public data available through August 28, 2026, and can be revised as new results emerge in energy, materials, robotics, and distributive policy. The second revised edition adds an analysis of the biological and cultural limits of abundance, explicitly separates evidence from speculation, replaces decorative formalizations with direct-language explanations, and consolidates subsections into longer argumentative units.